The Hong Kong Polytechnic University 香港理工大學
Department of Computing 電 子 計 算 學 系

Programme Requirement Document for

Bachelor of Science (Honours) Scheme

in

Computing and AI

電子計算及人工智能理學士組合課程

Programme Code: 61435

August 2023 (Updated in January 2024)

This Programme Requirement Document is applicable to students admitted in 2023/24

This Programme Requirement Document is subject to review and changes which the programme offering Faculty/Department can decide to make from time to time. Students will be informed of the changes as and when appropriate.

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Part 1: General Information

1.1 Introduction

1.1.1 Programme Title	:	Bachelor of Science (Honours) Scheme in Computing and AI
1.1.2 Programme Code	:	61435
1.1.3 Majors	:	Bachelor of Science (Honours) in Computer Science/ Computing (61435-FCS/61431-SYC) Bachelor of Science (Honours) in Enterprise Information Systems (61435-FIS/61431-SYS) Bachelor of Science (Honours) in Financial Technology and Artificial Intelligence (61435-FFT/61431-SYF)
1.1.4 Host Department	:	Department of Computing
1.1.5 Mode of Attendance	:	Full-time
1.1.6 Normal Duration	:	4 years (for normal year intake students) and 2 years (for senior year intake students)
1.1.7 Programme Structure	:	Credit-based
1.1.8 Final Awards	:	BSc (Honours) Degree in Computer Science (for normal year intake)/ Computing (for senior year intake) BSc (Honours) Degree in Enterprise Information Systems BSc (Honours) Degree in Financial Technology and Artificial Intelligence
1.1.9 Credit Requirements for Graduation	:	120 academic credits plus WIE training credits
1.1.10 Medium of Instruction	:	English (except for subjects taught in Chinese)
1.1.11 Implementation Date	:	September 2022

1.2 Entrance Requirements

Applicants must have satisfied the minimum university entrance requirement for Hong Kong Diploma of Secondary Education (HKDSE), or possess alternative equivalent qualifications such as Higher Diploma (HD), Associate Degree (AD), Hong Kong Advanced Level (HKAL), International Baccalaureate Diploma (IB), or General Certificate of Education A-level (GCEALE). Admission to the BSc (Hons) Scheme in Computing and AI is 100% Scheme-based. Applicants admitted into the BSc (Hons) Scheme in Computing and AI can *select* any of the constituent programmes before the end of first semester of year 2 study *without quota limitation*. Subject to approval from the corresponding departments and quota availability, admitted applicants can also opt for the two double major programmes in their study.

HKDSE applicants

(a) Level 2 or above in Mathematics and Liberal Studies; ***AND***

(b) Level 3 or above in English Language and Chinese Language; ***AND***

(c) Level 3 in 2 Other Elective subjects [can include Extended Modules of Mathematics (M1/M2)]

In addition to the general entrance requirements stated above, under BSc (Hons) Scheme in Computing and AI, the following subjects are given the highest subject weightings:

- English
- Mathematics
- Extended modules of Mathematics
- Economics
- Information and Communication Technology
- Business, Accounting and Financial Studies
- Business, Accounting and Financial Studies (Accounting)
- Business, Accounting and Financial Studies (Business Management)
- All single and combined science subjects

Non-HKDSE applicants

(d) An appropriate Diploma passed with credit or a Higher Certificate from a recognised institution; ***OR***

(e) An appropriate Associate Degree / Higher Diploma from a recognised institution (suitable candidates will be considered for advanced standing entry to the senior year curriculum (Section 2.2.2)).

The University accepts attainments in HKALE/ HKASLE/ GCEALE/ GCEASLE/IAL and IB for admission to its 4-year degree programmes. Applicants holding qualifications of IB or A-Level may be granted credit transfer upon admission. Applicants who have completed HD or AD may be considered for credit transfer and strong applicants may be considered for direct entry to Senior Year, i.e. Year 3, with an Advanced Standing Status.

Applicants may be required to meet additional criteria due to the evolution of the programmes.

1.3 University Mission

The Hong Kong Polytechnic University aspires to be a leading university that advances and transfers knowledge, and provides the best holistic education for the benefit of Hong Kong, the nation and the world. The PolyU's missions are:

A. To pursue impactful research that benefits the world.
B. To nurture critical thinkers, effective communicators, innovative problem solvers and socially responsible global citizens; and
C. To foster a University community in which all members can excel in their aspirations with a strong sense of belonging and pride.

1.4 Aims and Objectives of the Programmes

1.4.1 Background

In today's world, almost everybody has to use computers in some way. From the 2015 HKSAR Report on Manpower Projection to 2022, information and communications (at 1.4%) is one of the fastest growing economic sectors during the period from 2012 to 2022. This sector is also identified as having the greatest potential for driving the development of our economy. The programmes are developed to nurture talents and professional to support the fast evolution of the IT industry and advance the frontiers of knowledge for the future of Hong Kong.

CARE stands for the four pillars defining the Department's model of education for students' learning and development: **C**areer (i.e., career development in computing), **A**pplication (i.e., applying computing knowledge), **R**esearch (i.e., advancing computing knowledge), and **E**ntrepreneurship (i.e., realizing and innovating computing knowledge). CARE also reflects our care for students for their all-round development. Under the broad definition of IT, we distil the important elements comprising and supporting IT and structure them into different programmes under the BSc (Hons) Scheme in of Computing, with different focuses, under the common theme of Computing:

- The *BSc (Hons) in Computer Science/ Computing* programme (Computing) is the core Computing programme, with a focus on the applied nature of Computing and IT in the business setting, with strong practical elements in line with the mission of the University. The programme has a rather strong flavour on the applied systems development.

- The *BSc (Hons) in Enterprise Information Systems* programme (EIS) possesses a much stronger flavour in information systems and their management aspects. Unlike traditional Management Information Systems (MIS) programmes, our programme is designed in a more balanced manner, aiming to train graduates capable of performing some system development, as well as to manage them effectively in businesses as well as enterprises.

- The *BSc (Hons) in Financial Technology and Artificial Intelligence* programme (FinTech & AI) will focus on foundations of Internet technology, business and financial modelling, artificial intelligence, big data analytics, and information security.

1.4.2 Programme Aims

The programme aims of our programmes are listed below:

I. To produce graduates with good fundamental computing concepts, sound intellectual and practical skills, and the ability to creatively apply computing and related technologies to different information processing areas such as business, industry and public sectors.
II. To produce graduates who will be responsive to technical and human aspects of computing.
III. To produce graduates who possess an awareness of the ethical and social issues, who will be able to continue personal development and progress with the evolution in the computing discipline.
IV. To produce graduates who will be equipped with essential system development and system integration knowledge that can contribute in the global environment as well as to the development of Hong Kong and China.

Correlation between Programme Aims and University Mission:

Programme Aims	University Mission		
	A	B	C
I	X	X	X
II	X		X
III		X	X
IV	X	X	X

1.5 Institutional Learning Outcomes (ILOs)

It is PolyU's educational mission to nurture competent professionals who are also critical thinkers, effective communicators, innovative problem solvers, lifelong learners, and ethical leaders. The institutional learning outcomes for these attributes are provided as follows:

i. **Competent professional**: Graduates should be able to integrate and to apply in-depth discipline knowledge and specialised skills that are fundamental to functioning effectively as entry-level professional (professional competence); understand the global trends and opportunities related to their professions (global outlook); and demonstrate entrepreneurial spirit and skills in their work, including the discovery and use of opportunities, and experimentation with novel ideas (entrepreneurship)..
ii. **Critical thinker**: Graduates should be able to examine and critique the validity of information, arguments, and different viewpoints, and to reach sound judgments on the basis of credible evidence and logical reasoning.
iii. **Innovative problem solver**: Graduates should be able to identify and define problems in both professional and day-to-day contexts, and produce innovative solutions to solve problems.
iv. **Effective communicator**: Graduates should be able to comprehend and communicate effectively in English, and Chinese where appropriate, orally and in writing, in professional and day-to-day contexts.
v. **Lifelong learner**: Graduates should be able to recognise the need for continual learning and self-development, and be able to plan, manage and evaluate their own learning in pursuit of self-determined goals.
vi. **Ethical leader**: Graduates should have an understanding of leadership and be prepared to serve as a leader and a team player (leadership and teamwork); demonstrate self-

leadership and psychosocial competence in pursuing personal and professional development (intrapersonal competence); be capable of building and maintaining relationship and resolving conflicts in group work situations (interpersonal competence); and demonstrate ethical reasoning in professional and day-to-day contexts (ethical reasoning).

vii. **Socially responsible global citizen:** Graduates should have the capacity for understanding different cultures and social development needs in the local, national and global contexts (interest in culture and social development); and accept their responsibilities as professionals and citizens to society, their own nation and the world (social, national, and global responsibility).

1.6 Programme's Intended Learning Outcomes (PILOs)

The ***BSc (Hons) Scheme in Computing and AI*** is an outcome-oriented programme. Students are nurtured to become all-round professionals upon graduation. The learning outcomes apply to the whole programme, as well as to each individual subject. Upon finishing the constituent programmes, students would have achieved the full set of programme's intended learning outcomes.

1. Communicate effectively in Chinese and English at a level sufficient for project and system presentation and documentation.
2. Demonstrate a global outlook in factors of broad interests and professional responsibilities in culture and social development needs in the local, national and global contexts that can affect the way computing systems are developed and used, as well as to demonstrate entrepreneurial spirit and skills in their work.
3. Understand and value ethical issues in design and development of computing systems, and accept responsibilities as computing professionals and citizens to the society, own nation and the world.
4. Be responsive to and follow closely the advancement in computing and its impact with an attitude of continuous and lifelong learning.
5. Work together as a team, being able to build and maintain relationship and resolve conflicts in projects and other group work situations, and demonstrate self-leadership and psychosocial competence in pursuing personal and professional.
6. Apply related computing tools and software with an understanding of their effectiveness and limitations.
7. Understand and apply the fundamental knowledge and technology of computing for solving computing problems.
8. Think and reason critically in analysing a problem, identifying the requirements, and developing alternatives for its solution.
9. Achieve specialized outcomes with respect to constituent programmes (understand/apply):
   a. Computer Science/ Computing-
      Understand and apply computing and information technologies for solving practical problems as required by the industry.
   b. EIS-
      Understand the commercial needs from both computing and management perspectives and be able to solve information systems problems for businesses and enterprises.
   c. FinTech & AI-
      Understand and apply artificial intelligence and finance knowledge for solving FinTech-related problems.
10. Achieve specialized outcomes with respect to constituent programmes (design/implement/evaluate):

a. Computer Science/ Computing-
Be able to design, implement and evaluate computing systems and applications to fulfil required needs with social, environmental and other appropriate considerations.

b. EIS-
Be able to design for the implementation and evaluation of information systems and applications to fulfil required business and enterprise needs with social, environmental and other appropriate considerations.

c. FinTech & AI-
Be able to design, implement and evaluate FinTech-related systems and applications to fulfil required needs with social, environmental and other appropriate considerations.

Correlation between PILOs and ILOs:

ILOs / PILOs	i	ii	iii	iv	v	vi	vii
1				X			
2	X				X		X
3						X	X
4	X				X		X
5	X			X		X	X
6	X	X	X				
7		X	X				
8		X					
9	X		X				
10	X	X	X				

Correlation between PILOs and Programme Aims:

Programme Aims / PILOs	I	II	III	IV
1	X		X	X
2	X	X	X	X
3			X	
4	X	X		
5	X	X		
6	X	X		
7	X	X		X
8		X		X
9	X	X		X
10	X			

Part 2: Programme Structure and Curriculum Design

2.1 Programme Structure

Students in the BSc (Hons) Scheme in Computing and AI will complete any one of the nested programmes in four years. The first year (stage 1) is the basic or fundamental year for university study, with a good number of language and fundamental subjects and introductory subjects to the Computing discipline. Students will learn fundamental knowledge necessary for an all-round university education and also develop some appreciation in the area of computing. The second year (stage 2) is the broadening year, in which students will accumulate more knowledge in various disciplines, inclusive of computing discipline and relevant supporting disciplines, such as management and/or accounting. The fundamental knowledge acquired within the first two years is essential for students to build up their problem solving skills. The third year (stage 3) is the strengthening year, in which students will accumulate more knowledge in computing, strengthening them at an intermediate level. The subjects lay the necessary breadth for them to select subjects according to their interest in their final year. The final year (stage 4) is the specialization year, in which students will choose a number of subjects that he likes most that will give him more in-depth knowledge in the selected computing discipline. An important feature of BSc (Hons) Scheme in Computing and AI is that there is a good variety of elective subjects which students can select, as well as the flexibility of the programme structure. However, since the set of compulsory subjects start to vary after stage 1, students would need to ensure that they have completed the necessary set of subjects contributing to the award that they are seeking.

For all programmes, four foundational programming languages: **P**ython, **R**, **J**ava and **C** (pronounced as "**PR**o**J**e**C**t") will be covered for both object-oriented and procedural programming for system/application-oriented and data-oriented purposes as illustrated in the following figure. Furthermore, mathematics knowledge and skills will be strengthened as well (e.g. through programming exercises). This seeks to provide students with a strong foundation to learn other programming languages and advanced computing techniques. Upon learning these **PR**o**J**e**C**t languages, students would be expected to be able to carry out **O**perational, **E**nterprise, **T**echnological and **S**ystem **prOjEcTS**. The programmes will make effective use of project-based learning to develop students critical thinking and innovative thinking skills.

PRo**J**e**C**t Programming Pillars

	Procedural Programming	Object-oriented Programming
System/Application-oriented	**C**	**J**ava
Data-oriented	**P**ython and **R**	**P**ython

Students (except senior year intake students who are not allowed to take minors) can elect to take on additional subjects to form a minor. Normally, a minor consists of 6 subjects as stipulated by the minor-offering department. Students can make effective use of the 4 required free subjects (Level-0 subjects and training subjects will not be counted to fulfil free subject requirement for graduation purpose) which are included in the 120 credits graduation requirement to earn an additional minor. A student is allowed to study up to 150 credits without paying extra fee; the extra credits beyond 150 credits would be self-financed and be charged accordingly by the university.

To address the evolving development in IT and empower students with the competitive edge for their career development, a stream option is introduced with effect from 2019/20 intake cohort. A stream is a cluster of subjects that represent a significant body of knowledge in an important area. Depending on own interests and career plans, students can select elective subjects of the streams and obtain a certificate of recognition from COMP upon fulfilling the subject requirements for the stream concerned. This option is applicable for Computing/ Computer Science and EIS students only. The proposed streams and requirements are listed in section 2.2.1.

2.2 Normal Pattern of Study

The BSc (Hons) Scheme in Computing and AI curriculum is summarized in the table below. Students who take internship option shall follow the respective study pattern. Each subject would be of 3 credits unless stated otherwise. Normally, the subjects would be completed in four years (full-time), except for students with a significant number of credits transferred, such as Senior Year students. For LCR (Language and Communication Requirement) subjects, please refer to Section **2.5.1.1** for more details. For CAR (Cluster Area Requirement) subjects, please refer to Section **2.5.1.6**. For personal development subjects such as Service Learning and Healthy Lifestyle, please refer to Sections **2.5.1.5 and 2.5.1.7 respectively**. The study pattern for GUR subjects is indicative only. Students may take these subjects according to their own schedule.

2.2.1 a. Study Pattern for Normal Year Intake (Non-internship option)

	Award		
	BSc Computer Science	**BSc Enterprise Information System**	**BSc Financial Technology & Artificial Intelligence**
Stage 1	**Fundamental Year**		
Sem 1	Healthy Lifestyle **(0 credit)**	Healthy Lifestyle **(0 credit)**	Healthy Lifestyle **(0 credit)**
	COMP1002 Computational Thinking and Problem Solving **(4 credits)**	COMP1002 Computational Thinking and Problem Solving **(4 credits)**	COMP1002 Computational Thinking and Problem Solving **(4 credits)**
	COMP1004 Introduction to Artificial Intelligence and Data Analytics **(2 credits)**	COMP1004 Introduction to Artificial Intelligence and Data Analytics **(2 credits)**	COMP1004 Introduction to Artificial Intelligence and Data Analytics **(2 credits)**
	AMA1104 Introductory Probability **(2 credits)**	AMA1104 Introductory Probability **(2 credits)**	AMA1104 Introductory Probability **(2 credits)**
	MM1031 Introduction to Innovation and Entrepreneurship **(1 credit)**	MM1031 Introduction to Innovation and Entrepreneurship **(1 credit)**	MM1031 Introduction to Innovation and Entrepreneurship **(1 credit)**
	LCR English I	LCR English I	LCR English I
	Leadership Education and Development	Leadership Education and Development	Leadership Education and Development
Sem 2	COMP1011 Programming Fundamentals	COMP1011 Programming Fundamentals	COMP1011 Programming Fundamentals
	COMP1411 Introduction to Computer Systems	COMP1411 Introduction to Computer Systems	COMP1411 Introduction to Computer Systems
	COMP1433 Introduction to Data Analytics	COMP1433 Introduction to Data Analytics	COMP1433 Introduction to Data Analytics
	LCR Chinese	LCR Chinese	LCR Chinese
	LCR English II	LCR English II	LCR English II
	30 credits	*30 credits*	*30 credits*
Stage 2	**Broadening Year**		
Sem 1	COMP2011 Data Structures	COMP2011 Data Structures	COMP2011 Data Structures
	COMP2012 Discrete Mathematics	COMP2021 Object-oriented Programming	COMP2021 Object-oriented Programming
	COMP2021 Object-oriented Programming	COMP2121 E-business	COMP2411 Database Systems
	COMP2411 Database Systems	COMP2411 Database Systems	AF2108 Financial Accounting *or* AF2111 Accounting for Decision Making
	CAR 1	CAR 1	CAR 1
Sem 2	COMP2322 Computer Networking	COMP2322 Computer Networking	COMP2322 Computer Networking
	COMP2421 Computer Organization	AF2108 Financial Accounting *or* AF2111 Accounting for Decision Making	AF1605 Introduction to Economics[1]
	COMP2432 Operating Systems	MM2021 Management and Organization	AF3313 Business Finance
	Service Learning	Service Learning	Service Learning
	CAR 2	CAR 2	CAR 2
	30 credits	*30 credits*	*30 credits*
Stage 3	**Strengthening Year**		
Sem 1	COMP3211 Software Engineering	COMP3134 Business Intelligence and Customer Relationship Management	COMP3211 Software Engineering
	COMP3423 Human-Computer Interaction	COMP3211 Software Engineering	Finance Elective 1
	COMP3438 System Programming	COMP3423 Human-Computer Interaction	Finance Elective 2
	CLC3242P Chinese for Professional Communication in Computing **(2 credits)**	CLC3242P Chinese for Professional Communication in Computing **(2 credits)**	CLC3242P Chinese for Professional Communication in Computing **(2 credits)**
	CAR 3	CAR 3	CAR 3
	Free Subject 1	Free Subject 1	Free Subject 1

[1] AF1605 is also used to satisfy the CAR N requirement (i.e. AF1BN02). Thus, students will be required to take another 3-credit free elective subject to make up for the total credit requirement.

	Award		
	BSc Computer Science	**BSc Enterprise Information System**	**BSc Financial Technology & Artificial Intelligence**
Sem 2	COMP3334 Computer Systems Security	COMP3235 Software Project Management	COMP3334 Computer Systems Security
	COMP3511 Legal Aspects and Ethics of Computing **(2 credits)**	COMP3511 Legal Aspects and Ethics of Computing **(2 credits)**	COMP3511 Legal Aspects and Ethics of Computing **(2 credits)**
	COMP Elective 1	COMP Elective 1	COMP Elective 1
	CAR 4	CAR 4	Free Subject 2
	Free Subject 2	Free Subject 2	Free Subject 3
	31 credits	*31 credits*	*31 credits*
Stage 4	**Specialization Year**		
Sem 1	COMP4913 Capstone Project **(6 credits)**	COMP4913 Capstone Project **(6 credits)**	COMP4142 E-Payment and Crypto Currency
	ELC3524 Professional Communication for Computing Students **(2 credits)**	ELC3524 Professional Communication for Computing Students **(2 credits)**	COMP4431 Artificial Intelligence
	COMP Elective 2	COMP Elective 2	COMP4432 Machine Learning
	COMP Elective 3	COMP Elective 3	COMP4913 Capstone Project **(6 credits)**
	Free Subject 3	Free Subject 3	ELC3524 Professional Communication for Computing Students **(2 credits)**
Sem 2	COMP4913 Capstone Project **(0 credit)**	COMP4913 Capstone Project **(0 credit)**	COMP4531 Emerging Topics in FinTech
	COMP Elective 4	COMP Elective 4	COMP4913 Capstone Project **(0 credit)**
	COMP Elective 5	COMP Elective 5	COMP Elective 2
	COMP Elective 6	COMP Elective 6	Free Subject 4
	Free Subject 4	Free Subject 4	Free Subject 5
	29 credits	*29 credits*	*29 credits*
Total	**120 credits**	**120 credits**	**120 credits**

Note: The study pattern for GUR subjects is indicative only. Students may take these subjects according to their own schedule.

b. Study Pattern for Normal Year Intake (Internship option)

	Award		
	BSc Computer Science	**BSc Enterprise Information System**	**BSc Financial Technology & Artificial Intelligence**
Stage 1	**Fundamental Year**		
Sem 1	Healthy Lifestyle **(0 credit)**	Healthy Lifestyle **(0 credit)**	Healthy Lifestyle **(0 credit)**
	COMP1002 Computational Thinking and Problem Solving **(4 credits)**	COMP1002 Computational Thinking and Problem Solving **(4 credits)**	COMP1002 Computational Thinking and Problem Solving **(4 credits)**
	COMP1004 Introduction to Artificial Intelligence and Data Analytics **(2 credits)**	COMP1004 Introduction to Artificial Intelligence and Data Analytics **(2 credits)**	COMP1004 Introduction to Artificial Intelligence and Data Analytics **(2 credits)**
	AMA1104 Introductory Probability **(2 credits)**	AMA1104 Introductory Probability **(2 credits)**	AMA1104 Introductory Probability **(2 credits)**
	MM1031 Introduction to Innovation and Entrepreneurship **(1 credit)**	MM1031 Introduction to Innovation and Entrepreneurship **(1 credit)**	MM1031 Introduction to Innovation and Entrepreneurship **(1 credit)**
	LCR English I	LCR English I	LCR English I
	Leadership Education and Development	Leadership Education and Development	Leadership Education and Development
Sem 2	COMP1011 Programming Fundamentals	COMP1011 Programming Fundamentals	COMP1011 Programming Fundamentals
	COMP1411 Introduction to Computer Systems	COMP1411 Introduction to Computer Systems	COMP1411 Introduction to Computer Systems
	COMP1433 Introduction to Data Analytics	COMP1433 Introduction to Data Analytics	COMP1433 Introduction to Data Analytics
	LCR Chinese	LCR Chinese	LCR Chinese
	LCR English II	LCR English II	LCR English II
	30 credits	*30 credits*	*30 credits*

	Award		
	BSc Computer Science	**BSc Enterprise Information System**	**BSc Financial Technology & Artificial Intelligence**
Stage 2	**Broadening Year**		
Sem 1	COMP2011 Data Structures	COMP2011 Data Structures	COMP2011 Data Structures
	COMP2012 Discrete Mathematics	COMP2021 Object-oriented Programming	COMP2021 Object-oriented Programming
	COMP2021 Object-oriented Programming	COMP2121 E-business	COMP2411 Database Systems
	COMP2411 Database Systems	COMP2411 Database Systems	AF2108 Financial Accounting *or* AF2111 Accounting for Decision Making
	CAR 1	CAR 1	CAR 1
Sem 2	COMP2322 Computer Networking	COMP2322 Computer Networking	COMP2322 Computer Networking
	COMP2421 Computer Organization	AF2108 Financial Accounting *or* AF2111 Accounting for Decision Making	AF1605 Introduction to Economics[2]
	COMP2432 Operating Systems	MM2021 Management and Organization	AF3313 Business Finance
	Service Learning	Service Learning	Service Learning
	CAR 2	CAR 2	CAR 2
	30 credits	*30 credits*	*30 credits*
Stage 3	**Strengthening Year**		
Sem 1	COMP3211 Software Engineering	COMP3134 Business Intelligence and Customer Relationship Management	COMP3211 Software Engineering
	COMP3423 Human-Computer Interaction	COMP3211 Software Engineering	Finance Elective 1
	COMP3438 System Programming	COMP3423 Human-Computer Interaction	Finance Elective 2
	CLC3242P Chinese for Professional Communication in Computing **(2 credits)**	CLC3242P Chinese for Professional Communication in Computing **(2 credits)**	CLC3242P Chinese for Professional Communication in Computing **(2 credits)**
	CAR 3	CAR 3	CAR 3
	Free Subject 1	Free Subject 1	Free Subject 1
	Free Subject 2	Free Subject 2	Free Subject 2
Sem 2	COMP3334 Computer Systems Security	COMP3235 Software Project Management	COMP3334 Computer Systems Security
	COMP3511 Legal Aspects and Ethics of Computing **(2 credits)**	COMP3511 Legal Aspects and Ethics of Computing **(2 credits)**	COMP3511 Legal Aspects and Ethics of Computing **(2 credits)**
	COMP4001 Internship^	COMP4001 Internship^	COMP4001 Internship^
	25 credits	*25 credits*	*25 credits*
Stage 4	**Specialization Year**		
Sem 1	COMP4913 Capstone Project **(6 credits)**	COMP4913 Capstone Project **(6 credits)**	COMP4142 E-Payment and Crypto Currency
	ELC3524 Professional Communication for Computing Students **(2 credits)**	ELC3524 Professional Communication for Computing Students **(2 credits)**	COMP4431 Artificial Intelligence
	COMP Elective 1	COMP Elective 1	COMP4432 Machine Learning
	COMP Elective 2	COMP Elective 2	COMP4913 Capstone Project **(6 credits)**
	COMP Elective 3	COMP Elective 3	COMP Elective 1
	Free Subject 3	Free Subject 3	ELC3524 Professional Communication for Computing Students **(2 credits)**
Sem 2	COMP4913 Capstone Project **(0 credit)**	COMP4913 Capstone Project **(0 credit)**	COMP4531 Emerging Topics in FinTech
	COMP Elective 4	COMP Elective 4	COMP4913 Capstone Project **(0 credit)**

[2] AF1605 is also used to satisfy the CAR N requirement (i.e. AF1BN02). Thus, students will be required to take another 3-credit free elective subject to make up for the total credit requirement.

	Award		
	BSc Computer Science	**BSc Enterprise Information System**	**BSc Financial Technology & Artificial Intelligence**
	COMP Elective 5	COMP Elective 5	COMP Elective 2
	COMP Elective 6	COMP Elective 6	Free Subject 3
	CAR 4	CAR 4	Free Subject 4
	Free Subject 4	Free Subject 4	Free Subject 5
	35 credits	*35 credits*	*35 credits*
Total	**120 credits**	**120 credits**	**120 credits**

Note: The study pattern for GUR subjects is indicative only. Students may take these subjects according to their own schedule.
^COMP4001 is a 12-credit training subject.

The table below shows the list of electives for different programmes. Except for subjects marked as "NA" for a respective programme, students enjoy freedom in choosing among all the electives, if the pre-requisites are satisfied. Students may also choose to take Computing electives for the required free subjects if they desire to accumulate more knowledge in the Computing discipline. Note that the *list of electives is subject to change*, and *the subjects may not be offered every year*.

Elective Subject	**Award**		
	BScC	**BScEIS**	**BScFT**
AF3316 Investments	NA	NA	FE
AF3317 Risk Management	NA	NA	NA
AF4320 Corporate Finance	NA	NA	FE
AF4325 Wealth Management	NA	NA	NA
AMA3201 Computational Methods*	SE (CS)	NA	NA
AMA3231 Numerical Methods and Computing*	SE (CS)	NA	NA
AMA4850 Optimization Methods*	SE (CS)	NA	NA
AMA4688 Simulation*	SE (CS)	NA	NA
AMA3100 Number, Combinatorics and Statistics*	SE (CS)	NA	NA
COMP2012 Discrete Mathematics	COM	E	E
COMP2121 E-Business	E	COM	FE
COMP2421 Computer Organization	COM	E	NA
COMP2432 Operating Systems	COM	E	NA
COMP3011 Design and Analysis of Algorithms	SE (CS)	E	E
COMP3022 Algorithms Engineering	E	E	E
COMP3121 Social and Collaborative Computing	E	SE (DA)	E
COMP3122 Information Systems Development	E	E	E
COMP3131 Business and Information Systems Strategies	E	E	E
COMP3133 Chinese Language Processing	E	E	E
COMP3134 Business Intelligence and Customer Relationship Management	SE (AB)	COM	FE
COMP3233 Software Testing and Quality Assurance	E	E	NA
COMP3235 Software Project Management	E	COM	E
COMP3334 Computer Systems Security	COM	E	COM
COMP3335 Database Security	SE (SI)	NA	E
COMP3421 Web Application Design and Development	SE (SI)	E	E
COMP3422 Creative Digital Media Design	E	E	NA
COMP3423 Human-Computer Interaction	COM	COM	E
COMP3438 System Programming	COM	E	NA
COMP3531 IT Entrepreneurship	E	E	E
COMP3911 Service Learning and Civic Engagement in the Information Age	E	E	E
COMP4011 Theory of Computation	SE (CS)	E	NA
COMP4121 E-Commerce Technology and Applications	E	E	FE
COMP4122 Game Design and Development	E	E	NA
COMP4123 Business Process and Workflow Management	E	E	E
COMP4125 Operations Research and Logistics Management	E	SE (DA)	FE
COMP4127 Information Systems Audit and Control	E	E	E
COMP4133 Information Retrieval	E	SE (DA)	E
COMP4135 Knowledge and Information Management	SE (AB)	SE (DA)	E
COMP4136 Data Protection and Security	E	NA	E
COMP4142 E-Payment and Crypto Currency	NA	NA	COM

Elective Subject	Award		
	BScC	**BScEIS**	**BScFT**
COMP4146 Computational Finance	E	SE (DA)	FE
COMP4322 Internetworking Protocol, Software and Management	SE (SI)	E	NA
COMP4332 Mobile Security: Principles and Practice	SE (SI)	E	E
COMP4334 Principles and Practice of Internet Security	SE (SI)	E	E
COMP4342 Mobile Computing	SE (SI)	E	E
COMP4422 Computer Graphics	E	E	NA
COMP4423 Computer Vision	SE (AB)	E	E
COMP4424 Extended Reality: Theory and Practice	E	E	E
COMP4431 Artificial Intelligence	SE (AB)	E	COM
COMP4432 Machine Learning	SE (AB) / SE (CS)	E	COM
COMP4433 Data Mining and Data Warehousing	SE (AB)	SE (DA)	E
COMP4434 Big Data Analytics	SE (AB) / SE (CS)	SE (DA)	E
COMP4436 Artificial Intelligence of Things	SE (AB) / SE (SI)	E	E
COMP4442 Service and Cloud Computing	SE (SI)	E	E
COMP4512 Intellectual Property Protection and Management	NA	E	E
COMP4531 Emerging Topics in FinTech	NA	NA	COM
EIE3312 Linear Systems	SE (CS)	NA	NA
MM2021 Management and Organization	E	COM	NA
MM2711 Introduction to Marketing	NA	E	E

*Note: Students can only choose at most 2 of these AMA subjects.

Remarks:

COM – Compulsory
E – COMP Elective
FE – Finance Elective
NA – Not applicable
SE (AB) – Stream Elective (AI & Big Data)
SE (SI) – Stream Elective (Systems & Infrastructures)
SE (CS) – Stream Elective (Computer Science)
SE (DA) – Stream Elective (Data Analytics)

As mentioned in section 2.1, a stream option is introduced to Computing/ Computer Science and EIS students. The proposed streams and requirements are summarized in the following table:

Stream	**Computing/ Computer Science**			**EIS**
	AI and Big Data	**Systems and Infrastructures**	**Computer Science**	**Data Analytics**
Subject Requirements	*Complete 5 subjects from the list*			
	COMP3134 Business Intelligence and Customer Relationship Management	COMP3335 Database Security	COMP3011 Design and Analysis of Algorithms	COMP3121 Social and Collaborative Computing
	COMP4135 Knowledge and Information Management	COMP3421 Web Application Design and Development	COMP4011 Theory of Computation	COMP4125 Operations Research and Logistics Management
	COMP4423 Computer Vision	COMP4322 Internetworking Protocol, Software and Management	COMP4432 Machine Learning	COMP4133 Information Retrieval
	COMP4431 Artificial Intelligence	COMP4332 Mobile Security: Principles and Practice	COMP4434 Big Data Analytics	COMP4135 Knowledge and Information Management
	COMP4432 Machine Learning	COMP4334 Principles and Practice of Internet Security	AMA3201 Computational Methods*	COMP4146 Computational Finance
	COMP4433 Data Mining and Data Warehousing	COMP4342 Mobile Computing	AMA3231 Numerical Methods and Computing*	COMP4433 Data Mining and Data Warehousing
	COMP4434 Big Data Analytics	COMP4442 Service and Cloud Computing	AMA4850 Optimization Methods*	COMP4434 Big Data Analytics
			AMA4688 Simulation*	
	COMP4436 Artificial Intelligence of Things	COMP4436 Artificial Intelligence of Things	AMA3100 Number, Combinatorics and Statistics*	
			EIE3312 Linear Systems	

*Note: Students can only choose at most 2 of these AMA subjects

2.2.2 Study Pattern for Senior Year Intake (61431-SYC/SYS/SYF)

	Award		
	BSc Computing	**BSc Enterprise Information System**	**BSc FinTech & AI**
Sem 1, Year 1	COMP3211 Software Engineering	COMP2121 E-business	AF1605 Introduction to Economics
	COMP3423 Human-Computer Interaction	COMP3134 Business Intelligence and Customer Relationship Management	AF2108 Financial Accounting *or* AF2111 Accounting for Decision Making
	COMP Elective 1	COMP3211 Software Engineering	COMP3211 Software Engineering
	CAR M	COMP3423 Human-Computer Interaction	CAR M
	CLC3242P Chinese for Professional Communication in Computing **(2 credits)**	CLC3242P Chinese for Professional Communication in Computing **(2 credits)**	CLC3242P Chinese for Professional Communication in Computing **(2 credits)**
	GE Essential Components **(0 credit)**	GE Essential Components **(0 credit)**	GE Essential Components **(0 credit)**
Sem 2	COMP2421 Computer Organization	AF2108 Financial Accounting *or* AF2111 Accounting for Decision Making	AF3313 Business Finance
	COMP2432 Operating Systems	COMP3235 Software Project Management	COMP3334 Computer Systems Security
	COMP3334 Computer Systems Security	COMP3511 Legal Aspects and Ethics of Computing **(2 credits)**	COMP3511 Legal Aspects and Ethics of Computing **(2 credits)**
	COMP3511 Legal Aspects and Ethics of Computing **(2 credits)**	COMP Elective 1	COMP Elective 1
	CAR A – English Language	CAR A – English Language	CAR A – English Language
	Service Learning	Service Learning	Service Learning
	GE Essential Components **(0 credit)**	GE Essential Components **(0 credit)**	GE Essential Components **(0 credit)**
	31 credits	*31 credits*	*31 credits*
Summer	WIE (4 training credits)		
Sem 1	COMP3438 System Programming	COMP4913 Capstone Project	COMP4142 E-Payment and Crypto Currency
	COMP4913 Capstone Project	COMP Elective 2	COMP4431 Artificial Intelligence
	COMP Elective 2	COMP Elective 3	COMP4432 Machine Learning
	COMP Elective 3	CAR M	COMP4913 Capstone Project
	ELC3524 Professional Communication for Computing Students **(2 credits)**	ELC3524 Professional Communication for Computing Students **(2 credits)**	ELC3524 Professional Communication for Computing Students **(2 credits)**
Sem 2	COMP4913 Capstone Project	COMP4913 Capstone Project	COMP4531 Emerging Topics in FinTech
	COMP Elective 4	COMP Elective 4	COMP4913 Capstone Project
	COMP Elective 5	COMP Elective 5	Finance Elective 1
	COMP Elective 6	COMP Elective 6	Finance Elective 2
	COMP Elective 7	COMP Elective 7	COMP Elective 2
	29 credits	*29 credits*	*29 credits*
Total	**60 credits**	**60 credits**	**60 credits**

Remarks:

1. The senior year curriculum is articulating to that of the normal year cohort admitted 2 years ago. The study pattern shown above is aligned to the old programmes.
2. Those students not meeting the equivalent standard of the Undergraduate Degree LCR (based on their previous studies in AD/HD programme and their academic performance) will be required to take degree Language and Communication Requirement (LCR) subjects (up to 9 credits) on top of the above normal curriculum requirement. These additional LCR subjects will be included in the computation of GPAs.

The list of electives for senior year intake students is as listed in the table below.

Subject	Award		
	BScC	**BScEIS**	**BScFT**
AF3316 Investments	NA	NA	FE
AF4320 Corporate Finance	NA	NA	FE
AMA3201 Computational Methods*	SE (CS)	NA	NA
AMA3231 Numerical Methods and Computing*	SE (CS)	NA	NA
AMA4850 Optimization Methods*	SE (CS)	NA	NA
AMA4688 Simulation*	SE (CS)	NA	NA
AMA3100 Number, Combinatorics and Statistics*	SE (CS)	NA	NA
COMP3011 Design and Analysis of Algorithms	SE(CS)	E	E
COMP3022 Algorithms Engineering	E	E	E
COMP3121 Social and Collaborative Computing	E	SE (DA)	E
COMP3122 Information Systems Development	E	E	E
COMP3131 Business and Information Systems Strategies	E	E	E
COMP3133 Chinese Language Processing	E	E	E
COMP3134 Business Intelligence and Customer Relationship Management	SE (AB)	COM	FE
COMP3233 Software Testing and Quality Assurance	E	E	NA
COMP3235 Software Project Management	E	COM	E
COMP3334 Computer Systems Security	COM	E	COM
COMP3335 Database Security	SE (SI)	NA	E
COMP3421 Web Application Design and Development	SE (SI)	E	E
COMP3422 Creative Digital Media Design	E	E	NA
COMP3423 Human-Computer Interaction	COM	COM	NA
COMP3438 System Programming	COM	E	NA
COMP3531 IT Entrepreneurship	E	E	E
COMP3911 Service Learning and Civic Engagement in the Information Age	E	E	E
COMP4011 Theory of Computation	SE (CS)	E	NA
COMP4121 E-Commerce Technology and Applications	E	E	FE
COMP4122 Game Design and Development	E	E	NA
COMP4123 Business Process and Workflow Management	E	E	E
COMP4125 Operations Research and Logistics Management	E	SE (DA)	FE
COMP4127 Information Systems Audit and Control	E	E	E
COMP4133 Information Retrieval	E	SE (DA)	E
COMP4135 Knowledge and Information Management	SE (AB)	SE (DA)	E
COMP4136 Data Protection and Security	E	NA	E
COMP4142 E-Payment and Crypto Currency	NA	NA	COM
COMP4146 Computational Finance	E	SE (DA)	FE
COMP4322 Internetworking Protocol, Software and Management	SE (SI)	E	NA
COMP4332 Mobile Security: Principles and Practice	SE (SI)	E	E
COMP4334 Principles and Practice of Internet Security	SE (SI)	E	E
COMP4342 Mobile Computing	SE (SI)	E	E
COMP4422 Computer Graphics	E	E	NA
COMP4423 Computer Vision	SE (AB)	E	E
COMP4424 Extended Reality: Theory and Practice	E	E	E
COMP4431 Artificial Intelligence	SE (AB)	E	COM
COMP4432 Machine Learning	SE (AB)/ SE (CS)	E	COM
COMP4433 Data Mining and Data Warehousing	SE (AB)	SE (DA)	E
COMP4434 Big Data Analytics	SE (AB)/ SE (CS)	SE (DA)	E
COMP4436 Artificial Intelligence of Things	SE (AB) / SE (SI)	E	E
COMP4442 Service and Cloud Computing	SE (SI)	E	E
COMP4512 Intellectual Property Protection and Management	NA	E	E
COMP4531 Emerging Topics in FinTech	NA	NA	COM
EIE3312 Linear Systems	SE (CS)	NA	NA

*Note: Students can only choose at most 2 of these AMA subjects.

Remarks:

COM – Compulsory

E – COMP Elective

FE – Finance Elective
NA – Not applicable
SE (AB) – Stream Elective (AI & Big Data)
SE (SI) – Stream Elective (Systems & Infrastructure)
SE (CS) – Stream Elective (Computer Science)
SE (DA) – Stream Elective (Data Analytics)

COMP subject pre-requisites, weighting and other information are listed:

Subject Code	Subject Title	Credit	Weight	Pre-requisite	Contact Hours		Assessment	
					Lecture	T/L/S	CA	Exam
COMP1002	Computational Thinking and Problem Solving	4	2	-	39	13	55	45
COMP1004	Introduction to Artificial Intelligence and Data Analytics	2	2	-	23	0	100	0
COMP1011	Programming Fundamentals	3	2	-	39	13	65	35
COMP1411	Introduction to Computer Systems	3	2	-	26	13	55	45
COMP1433	Introduction to Data Analytics	3	2	-	39	0	45	55
COMP2011	Data Structures	3	2	COMP1011	39	13	60	40
COMP2012	Discrete Mathematics	3	2	-	26	13	60	40
COMP2021	Object-oriented Programming	3	2	COMP1011	39	13	60	40
COMP2121	E-Business	3	2	-	26	13	60	40
COMP2322	Computer Networking	3	2	COMP1011	39	13	55	45
COMP2411	Database Systems	3	2	COMP1011	39	13	55	45
COMP2421	Computer Organization	3	2	-	39	13	55	45
COMP2432	Operating Systems	3	2	COMP2011	39	13	55	45
COMP3011	Design and Analysis of Algorithms	3	3	COMP2011	26	13	60	40
COMP3022	Algorithms Engineering	3	3	COMP1011, COMP1411, COMP2011	26	13	60	40
COMP3121	Social and Collaborative Computing	3	3	-	39	0	60	40
COMP3122	Information Systems Development	3	3	COMP2411	39	0	70	30
COMP3131	Business and Information Systems Strategies	3	3	-	39	0	60	40
COMP3133	Chinese Language Processing	3	3	COMP1011	26	13	55	45
COMP3134	Business Intelligence and Customer Relationship Management	3	3	-	39	0	55	45
COMP3211	Software Engineering	3	3	COMP2011	39	0	60	40
COMP3233	Software Testing and Quality Assurance	3	3	COMP3211	39	0	55	45
COMP3235	Software Project Management	3	3	-	36	3	55	45
COMP3334	Computer Systems Security	3	3	COMP2432	39	0	60	40
COMP3335	Database Security	3	3	COMP2411	39	0	55	45
COMP3421	Web Application Design and Development	3	3	COMP1011	26	13	55	45
COMP3422	Creative Digital Media Design	3	3	COMP2011	26	13	55	45
COMP3423	Human-Computer Interaction	3	3	COMP1011	39	0	60	40
COMP3438	System Programming	3	3	COMP2432	39	13	55	45
COMP3511	Legal Aspects and Ethics of Computing	2	3	-	26	0	100	0
COMP3531	IT Entrepreneurship	3	3	-	39	0	65	35
COMP3911	Service Learning and Civic Engagement in the Information Age	3	3	-	28	10	100	0
COMP4011	Theory of Computation	3	3	COMP3011	26	13	60	40
COMP4121	E-Commerce Technology and Applications	3	3	COMP2411	39	0	55	45
COMP4122	Game Design and Development	3	3	COMP2011	26	13	55	45
COMP4123	Business Process and Workflow Management	3	3	-	39	0	55	45
COMP4125	Operations Research and Logistics Management	3	3	COMP1011, COMP2011	39	0	60	40
COMP4127	Information Systems Audit and Control	3	3	-	39	0	55	45
COMP4133	Information Retrieval	3	3	COMP2011, COMP2411	39	0	55	45
COMP4135	Knowledge and Information Management	3	3	-	39	0	55	45
COMP4136	Data Protection and Security	3	3	COMP1011	39	0	55	45
COMP4142	E-Payment and Cryptocurrency	3	3	COMP3334	39	0	55	45
COMP4146	Computational Finance	3	3	-	39	0	60	40
COMP4322	Internetworking Protocol, Software and Management	3	3	COMP2322	39	0	50	50
COMP4332	Mobile Security: Principles and	3	3	COMP1011,	26	13	55	45

Subject Code	Subject Title	Credit	Weight	Pre-requisite	Contact Hours		Assessment	
					Lecture	T/L/S	CA	Exam
	Practice			COMP2322, COMP2432/COMP1411, COMP3334				
COMP4334	Principles and Practice of Internet Security	3	3	COMP3334	39	0	60	40
COMP4342	Mobile Computing	3	3	COMP2411, COMP2432/COMP1411	26	13	55	45
COMP4422	Computer Graphics	3	3	COMP2011	26	13	60	40
COMP4423	Computer Vision	3	3	COMP2011/ EIE2106	26	13	55	45
COMP4424	Extended Reality: Theory and Practice	3	3	COMP2222/ COMP3423	39	0	70	30
COMP4431	Artificial Intelligence	3	3	COMP2011	39	0	55	45
COMP4432	Machine Learning	3	3	-	39	0	55	45
COMP4433	Data Mining and Data Warehousing	3	3	COMP2411	39	0	55	45
COMP4434	Big Data Analytics	3	3	AMA1104, COMP1011, COMP2011, COMP2411	26	13	60	40
COMP4436	Artificial Intelligence of Things	3	3	COMP1011	26	13	60	40
COMP4442	Service and Cloud Computing	3	3	COMP2322	39	0	55	45
COMP4512	Intellectual Property Protection and Management	3	3	-	39	0	55	45
COMP4531	Emerging Topics in FinTech	3	3	-	26	0	55	45
COMP4913	Capstone Project	6	3	-	0	0	100	0

2.3 Curriculum Map

The curriculum map indicates which subject contributes to the programme's intended learning outcomes, for which the subject teach (T), give students practice (P) and measure (M) the programme outcomes.

Programme Outcome / Subject	1	2	3	4	5	6	7	8	9			10		
									Comp	EIS	FTAI	Comp	EIS	FTAI
AF1605 Introduction to Economics	P	P							TP	TP	TP			
AF2108 Financial Accounting	P													
AF2111 Accounting for Decision Making	P													
AF3313 Business Finance	P	P		P				TP						
AF3316 Investments	P													
AF3317 Risk Management	P	P		P				TP						
AF4320 Corporate Finance	P		P											
AF4325 Wealth Management	P	P		P				TP						
AMA1104 Introductory Probability and Statistics								TP						
AMA3201 Computational Methods*						TP	TP	TP						
AMA3231 Numerical Methods and Computing*						TP	TP	TP						
AMA4850 Optimization Methods*							TP	TP						
AMA4688 Simulation*						TP		TP	TP					
AMA3100 Number, Combinatorics and Statistics*							TP	TP	TP					
APSS1L01 Tomorrow's Leader#			T	T	TP									
CLC3242P Chinese for Professional Communications in Computing	TPM													
COMP1002 Computational Thinking and Problem Solving					P	TP	TP	TP						
COMP1004 Introduction to Artificial Intelligence and Data Analytics						TPM	TPM	TPM	TP		TP			
COMP1011 Programming Fundamentals					P	TPM	TPM	TPM						
COMP1411 Introduction to Computer Systems						TP	TP	TP						
COMP1433 Introduction to Data Analytics		TP				TP	TP	TP						
COMP2011 Data Structures						TPM	TPM	TPM						
COMP2012 Discrete Mathematics						TP	TP	TP						
COMP2021 Object-oriented Programming					P	TPM	TPM	TPM						
COMP2121 E-Business	P	TP	T			TP	TP	TP	TP	TP	TP	TP	TP	TP
COMP2322 Computer Networking	P				P	TPM	TPM	TPM	TPM	TPM	TPM	TPM	TPM	TPM
COMP2411 Database Systems	TPM			TP	TPM	TPM	TPM	TPM	TPM	TPM	TPM	TPM	TPM	TPM
COMP2421 Computer Organization				T		TP	TP	TP	TP			TP		
COMP2432 Operating Systems	P				P	TPM	TPM	TPM	TP			TP		
COMP3011 Design and Analysis of Algorithms						TPM	TPM	TPM						
COMP3022 Algorithms Engineering	P					TPM	TPM	TPM						
COMP3121 Social and Collaborative Computing	PM	TPM	TPM		P				TP	TP	TP	TP	TP	TP
COMP3122 Information Systems Development	P	TP			P	TPM	TPM	TPM	TP	TP	TP	TP	TP	TP
COMP3131 Business and Information Systems Strategies	P	TP	T		P	TPM		TPM	TP	TP	TP	TP	TP	TP

Programme Outcome / Subject	1	2	3	4	5	6	7	8	9			10		
									Comp	EIS	FTAI	Comp	EIS	FTAI
COMP3133 Chinese Language Processing		TPM				TPM	TPM	TPM	TP	TP	TP	TP	TP	TP
COMP3134 Business Intelligence and Customer Relationship Management	P	TP	TP		P	TPM	TPM	TPM	TP	TP	TP	TP	TP	TP
COMP3211 Software Engineering	TP				P	TPM	TPM	TPM	TPM	TPM	TPM	TPM	TPM	TPM
COMP3233 Software Testing and Quality Assurance	P				P	TPM	TPM	TPM	TPM	TPM	TPM	TPM	TPM	TPM
COMP3235 Software Project Management	PM		TP		PM	PM	TP	PM	TPM	TPM	TPM	TPM	TPM	TPM
COMP3334 Computer Systems Security	PM				P	TPM	TPM	TPM	TPM		TPM	TPM		TPM
COMP3335 Database Security					TP	TP	TP	TP	TP			TP		
COMP3421 Web Application Design and Development	TP			TP	P	TPM	TPM	TPM	TP	TP	TP	TP	TP	TP
COMP3422 Creative Digital Media Design		T		T	P	TP	TPM	TP	TP			TP		
COMP3423 Human-Computer Interaction	PM	P	TPM		P	TPM	TPM	TPM	TP	TP	TP	TP	TP	TP
COMP3438 System Programming					P	TP	TPM	TP	TP			TP		
COMP3511 Legal Aspects and Ethics of Computing	PM	TPM	TPM	TPM	PM				PM	PM	PM	PM	PM	PM
COMP3531 IT Entrepreneurship	PM	TPM	TPM	T	P				TP	TPM		TP	TPM	
COMP3911 Service Learning and Civic Engagement in the Information Age	TP		TPM		PM	TPM		TPM						
COMP4011 Theory of Computation						TPM	TPM	TPM						
COMP4121 E-Commerce Technology and Applications	P	T	T		P	TPM	TPM	TPM	PM	PM	PM	PM	PM	PM
COMP4122 Game Design and Development	PM	TPM	TPM	T	PM	TPM	TPM	TPM	TPM			TPM		
COMP4123 Business Process and Workflow Management	P	T	T	T	P	PM	TP	PM		TPM			TPM	
COMP4125 Operations Research and Logistics Management	TP				P	TPM	TP	TPM		TPM			TPM	
COMP4127 Information Systems Audit and Control	P	TP	TPM	TP	P	TPM		TPM	TPM	TPM	TPM	TPM	TPM	TPM
COMP4133 Information Retrieval	P				P	TPM	TPM	TPM	TP			TP		
COMP4135 Knowledge and Information Management	PM	T	T		P	TPM	TPM	TPM		TPM			TPM	
COMP4136 Data Protection and Security		TM	TM	TPM	P	TPM	TPM	TPM	P	P	P	P	P	P
COMP4142 E-Payment and Cryptocurrency	P				P	TPM	TPM	TPM		TPM	TPM		TPM	TPM
COMP4146 Computational Finance	P			T	P	TPM	TPM	TPM	TP	TP	TP	TP	TP	TP
COMP4322 Internetworking Protocol, Software and Management	P			T	PM	TPM	TPM	TPM	PM			PM		
COMP4332 Mobile Security: Principles and Practice				T	PM	TPM	TPM	TPM	TPM			TPM		
COMP4334 Principles and Practice of Internet Security	PM				PM	TPM	TPM	TPM	TPM			TPM		
COMP4342 Mobile Computing	P			T	P	TPM	TPM	TPM	PM			PM		
COMP4422 Computer Graphics				TP	P	TPM	TPM	TPM	TPM			TPM		
COMP4423 Computer Vision				T	P	TPM	TPM	TPM	TP			TP		
COMP4424 Extended Reality: Theory and Practice		P			PM	TPM	TPM	TPM	TPM					
COMP4431 Artificial Intelligence					P	TPM	TPM	TPM	TPM		TPM	TPM		TPM
COMP4432 Machine Learning						TPM	TPM	TPM	TP	TP	TP			
COMP4433 Data Mining and Data Warehousing	P					TPM	TPM	TPM	TPM	TPM	TPM	TPM	TPM	TPM
COMP4434 Big Data Analytics						TPM	TPM	TPM	TPM	TPM	TPM	TPM	TPM	TPM
COMP4436 Artificial Intelligence of Things	P	TPM		TPM	TPM	TPM	TPM	TPM	TPM	TPM	TPM	T	T	T
COMP4442 Service and Cloud Computing	P	TP	TP	P	PM	TPM	TPM	TPM	PM			PM		

Programme Outcome / Subject	1	2	3	4	5	6	7	8	9			10		
									Comp	EIS	FTAI	Comp	EIS	FTAI
COMP4512 Intellectual Property Protection and Management	P		TP	T		TP	TP	TP		TP	TP		TP	TP
COMP4531 Emerging Topics in FinTech	P	TP		TP	P	TPM	TPM	TPM	TPM	TPM	TPM	TPM	TPM	TPM
COMP4913 Capstone Project	PM	PM		PM	PM	PM	PM	PM	PM	PM	PM	PM	PM	PM
EIE3312 Linear Systems							TP	TP						
ELC3524 Professional Communication for Computing Students	TPM													
MM2021 Management and Organization			P											
MM2711 Introduction to Marketing	P	P	P											

* Students can only choose at most 2 of these AMA subjects.
# Students can take other University's approved leadership subject.

2.4 Work Integrated Education

The Hong Kong Polytechnic University is promoting Work Integrated Education (WIE) as one of the important mechanisms to improve the all-roundedness of students. Mandatory WIE should be in alignment with PolyU's strategic goal of providing value-added education leading to the development of all-round students with professional competence. This requires that the WIE activities should aim to achieve learning outcomes both along professional knowledge and skills and all-roundedness.

Mandatory WIE activities should be *structured* as follows:

- There should be intended learning outcomes set for the workplace learning.
- Work experience should be purposefully designed to provide intentional learning aimed at the attainment of the intended outcomes, instead of leaving learning to occur incidentally as a side effect of work.
- Appropriate mechanisms of support provided by PolyU and workplace supervisors should be devised to ensure that effective learning does take place.

Mandatory WIE activities should be *measured* in terms of the following:

- Students should be required to document their workplace learning experience using instruments appropriate for demonstrating attainment of WIE learning outcomes, for example, reports, portfolios, etc.
- Assessment of the attainment of intended learning outcomes and the provision of student feedback should be built in.

Mandatory WIE activities are credit-bearing, but they are not included into the 120 academic credits required for graduation. The WIE components will ***NOT*** be counted towards GPA calculation. Students will earn ***one credit*** for the completion of every ***two weeks*** of WIE activities.

Students in the BSc (Hons) Scheme in Computing and AI must accumulate at least 4 WIE credits before they can graduate.

The WIE supervisor of a student will be in charge of monitoring the progress of the student, who must actively report his/her progress and consult with industrial supervisor to align his/her work with the agreed upon learning outcomes (as stipulated in Section **2.4.1** and **2.4.2**). WIE supervisor will provide advice and guidelines, and can interact with industrial supervisor to ensure that learning outcomes can be attained.

2.4.1 Intended Learning Outcomes of the WIE Component

The WIE component complements the student's all-round learning. Upon completion of the WIE, a student should be able to know better how the industry works. Therefore, the WIE should offer students a chance *to apply his technical skills in solving real-life problems*. In general, the student should improve on *problem solving skills* and *critical thinking*, in acquiring better *global outlook* in industrial standard and practices. More importantly, *team spirit*, *communication* and *interpersonal skills* should be developed. Depending on the actual job details, the exact learning outcomes may vary and is to be set by student and endorsed by an academic staff serving as a ***WIE supervisor*** at the beginning of the WIE undertaking.

2.4.2 Structure of the WIE Component

Students must inform the Department in advance to set up the learning outcomes as well as to confirm the WIE job period, which will vary from case to case. Inappropriate jobs

will be rejected and will not count towards WIE. A student can take summer or term-break WIE job preferably after year one and should complete at least 312 hours for obtaining 4 WIE credits (one WIE credit corresponds to 78 hours of approved work) before he can graduate. A final report with assessment is needed for the passing grade and granting of WIE credits. This is considered as a formal subject COMP3000.

As an alternative to COMP3000, an optional internship (COMP4001) is also available. During the internship period (Year 3 Semester 2 and Summer), students can take two or more evening subjects. This option allows students to complete their study in four years while gaining up to 32 weeks of working or industrial experience.

2.4.3 Assessment of the WIE Component

WIE is assessed throughout the duration of the job period. Each WIE registration must be accompanied with a request *student* detailing the job descriptions. The expected learning outcomes should also be specified in the request, which upon endorsement by the department, will be used as an evaluation yardstick against the student during the job period. Students must submit a *written report* detailing his/her achievement during the WIE job period, with a *reflection on the learning outcomes* attained. A passing grade is awarded only when the learning outcomes are basically met, as judged by the WIE supervisor through evidence provided by the student. WIE credits will not be counted towards the Grade Point Average (GPA).

More details of COMP WIE could be viewed at https://wie.comp.polyu.edu.hk/.

2.5 University Graduation Requirements for 4-year Full-time Undergraduate Degree offered from 2012/13 onward

All candidates qualifying for a 4-year Full-time Undergraduate Degree offered from 2012/13 onward must meet:

1. the University Graduation Requirements, and
2. the specific graduation requirements of their chosen programme of study (Majors and Minors).

The minimum University Graduation Requirements are explained in the sections below. For the graduation requirements of specific programmes of study (majors and minors), candidates should refer to the relevant section of the Programme Requirement Document.

2.5.1 University Graduation Requirements

To be eligible for a PolyU Bachelor's Degree (including the senior year intake) under the 4-year full-time undergraduate curriculum, a student must:

1. Complete successfully a minimum of 120 credits (60 credits for senior year intake).
2. Earn a cumulative GPA of 1.70 or above at graduation.
3. Complete successfully the mandatory Work-Integrated Education (WIE) component as specified by their programme/major.
4. Satisfy the following GUR requirements:

Requirement	Credits required for 4-yr Full-time students	Credits required for 2-yr Full-time Senior Year Intake students
(a) Language and Communication Requirements	9 credits	- (only except those not meeting the equivalent standard of the Undergraduate Degree LCR)
(b) Artificial Intelligent and Data Analytics (AIDA)	2 credits	-
(c) Innovation and Entrepreneurship (IE)	1 credit	-
(d) Leadership Education and Development	3 credits	-
(e) Service-Learning	3 credits	3 credits
(f) Cluster Areas Requirement (CAR)	12 credits (3 credits from each of the 4 cluster areas)	6 credits (3 credits from CAR(A)* and 3 credits from CAR(M), * Students are required to take a specially designed CAR(A) with embedded English Reading and Writing Requirements)
(g) Healthy Lifestyle	Non-credit bearing	-
(h) Essential Components of GE	-	Non-credit bearing
	Total = 30 credits	***Total = 9 credits***

5. Satisfy the residential requirement (i.e. must complete at least 1/3 of the normal credit requirement) for the award he/she is currently enrolled, unless the professional bodies concerned stipulate otherwise.

2.5.1.1 Language and Communication Requirement (LCR)

The overall aim of the requirements is to facilitate students in satisfying the University's objectives of biliteracy and/or trilingualism in terms of their:

1) general language proficiency
2) language knowledge and skills necessary for effective study at university level
3) literacy skills (reading and writing)
4) language knowledge and skills necessary for entry into the Broad Discipline's discourse communities

These four aspects are addressed in the four major components of the overall English and Chinese language requirements, which must be fulfilled as part of the graduation requirements:

1) LCR in English (6 credits) and Chinese (3 credits)
2) Writing Requirement (W)
3) Reading Requirement (R)
4) Discipline-Specific Language Requirements (2 credits each in English & Chinese)

English

All undergraduate students must successfully complete <u>two</u> 3-credit English language subjects as stipulated by the University (Table 1). These subjects are designed to suit students' different levels of English language proficiency at entry, as determined by their Hong Kong Diploma of Secondary Education (HKDSE) score or the English Language Centre (ELC) entry assessment (when no HKDSE score is available, e.g. in the case of non-local students).

Students entering the University with specified attainment grades in certain public examinations can be given credit transfer or exemption for one or both LCR English subjects.

Table 1. English LCR subjects (each 3 credits)

English Language competence level/ Subject	*Practical English for University Studies*	*English for University Studies*	*Any LCR Proficient level elective subject in English (Table 2)*
HKDSE Level 4 and above or equivalent	--	Subject 1	Subject 2
HKDSE Level 3 or equivalent	Subject 1	Subject 2	--

Table 2. Proficient level subjects for DSE Level 4 students and above (or equivalent) (each 3 credits)

LCR Proficient level elective subjects	Advanced English for University Studies	3 credits each
	Advanced English Reading and Writing Skills	
	Persuasive Communication	
	English in Literature and Film	

Chinese

All undergraduate students are required to successfully complete one 3-credit Chinese language subject as stipulated by the University, according to their Chinese language proficiency level (Table 3).

Students who are non-Chinese speakers (NCS), or whose Chinese standards are at junior secondary level or below, will by default be exempted from the Discipline-Specific Chinese Language requirement and CAR-Chinese Reading and Writing requirements. They can take a replacement subject of any level to make up for credit requirement. However, they will still be required to take one Chinese LCR subject specially designed to suit their language background and entry standard as shown in Table 4.

Table 3. Chinese LCR subjects

Categories of students	Required subject
For Chinese speaking students	University Chinese (Cantonese or Putonghua version)
For non-Chinese speakers or students whose Chinese standards are at junior secondary level or below	One subject from Table 4 below

Table 4. Chinese LCR Subjects for non-Chinese speakers or students whose Chinese standards are at junior secondary level or below

Subject (each 3 credits)	Pre-requisite/exclusion
Chinese I (for non-Chinese speaking students)	For non-Chinese speaking students at beginners' level
Chinese II (for non-Chinese speaking students)	• For non-Chinese speaking students; and • Students who have completed Chinese I or equivalent
Chinese III (for non-Chinese speaking students)	• For non-Chinese speaking students at higher competence levels; and • Students who have completed Chinese II or equivalent
Chinese IV (for Non-Chinese	• For non-Chinese students at intermediate competence

speaking students)	levels; and • Students who have completed Chinese III or equivalent
Chinese Literature – Linguistics and Cultural Perspectives (for non-Chinese speaking students)	• For non-Chinese speaking students at higher competence levels

Students who have obtained verified qualifications or certain results in some public examinations [e.g. HKDSE, HKALE, JEE, GSAT(Taiwan)] may be granted credit transfer/exemption for the Chinese LCR subject.

Writing Requirement

In addition to the LCR in English and Chinese explained above, all students must also, among the Cluster Areas Requirement (CAR) subjects they take (see section 2.5.1.6 below), pass one subject that includes the requirement for a substantial piece of writing in English and one subject with the requirement for a substantial piece of writing in Chinese.

Reading Requirement

All students must, among the CAR subjects they take, pass one subject that includes the requirement for the reading of an extensive text in English and one subject with the requirement for the reading of an extensive text in Chinese.

A list of approved CAR subjects for meeting the Writing Requirement (with a "W" designation) and for meeting the Reading Requirement (with an "R" designation) is shown at https://www.polyu.edu.hk/ogur/GURSubjects/.

In addition to the LCR and Reading and Writing Requirements, students also have to complete 4 credits of discipline-specific language requirements (2 credits in English and 2 credits in Chinese) as specified in the curriculum requirements of their Major.

2.5.1.2 Artificial Intelligence and Data Analytics (AIDA)

All students must successfully complete, normally in their first year of study, one 2-credit AIDA subject, which is COMP1004 Introduction to Artificial Intelligence and Data Analytics.

2.5.1.3 Innovation and Entrepreneurship (IE)

All students must successfully complete, normally in their first year of study, one 1-credit IE subject, which is MM1031 Introduction to Innovation and Entrepreneurship.

2.5.1.4 Leadership Education and Development

All students must successfully complete one 3-credit subject in the area of Leadership Education and Development, which is designed to enable students to:

(1) understand and integrate theories, research and concepts on the qualities (particularly intra-personal and interpersonal qualities) of effective leaders in the Chinese context,

(2) develop greater self-awareness and a better understanding of themselves,

(3) acquire interpersonal skills essential for functioning as an effective leader,

(4) develop self-reflection skills in their learning, and

(5) recognise the importance of the active pursuit of knowledge on an intra-personal and interpersonal level, and its relationship to leadership qualities.

A list of designated subjects for meeting the leadership education and development requirement is available at https://www.polyu.edu.hk/ogur/GURSubjects/.

2.5.1.5 Service-Learning

All students must successfully complete one 3-credit subject designated to meet the Service-Learning requirement, in which they are required to:

(1) participate in substantial community service or civic engagement activities that will benefit the service users or the community at large in a meaningful way,

(2) apply the knowledge and skills acquired from their Major or other learning experiences at the University to the community service activities, and

(3) reflect on their service learning experience in order to link theory with practice for the development of a stronger sense of ethical, social and national responsibility.

These subjects may take the form of:

- An open-to-all GUR service-learning subject
- A GUR Service-Learning subject targeting a particular student group (e.g. a Broad Discipline), or
- A customised DSR subject (core or elective) within the Major/Minor, with all the required features and components to meet the Service-Learning requirement.

Students who have satisfied the Service-Learning requirement via a customised DSR subject will be required to take another 3-credit subject to make up for the total credit requirement.

A list of designated subjects for meeting the Service-Learning requirement is available at https://www.polyu.edu.hk/ogur/GURSubjects/. The Department of Computing offers an elective subject COMP3911 and this subject could also be used to satisfy the Service-Learning requirement. Students who have satisfied the Service-Learning requirement via COMP3911 will be required to take another 3-credit subject to make up for the total credit requirement.

2.5.1.6 Cluster Area Requirement (CAR)

To expand students' intellectual capacity beyond their disciplinary domain and to enable them to tackle professional and global issues from a multidisciplinary perspective, students are required to successfully complete at least one 3-credit subject in each of the following four Cluster Areas:

(1) Human Nature, Relations and Development (CAR A)
(2) Science, Technology and Environment (CAR D)
(3) Chinese History and Culture (CAR M)
(4) Cultures, Organizations, Societies and Globalisation (CAR N)

A list of CAR subjects under each of the four Cluster Areas is available at https://www.polyu.edu.hk/ogur/GURSubjects/

For senior year students, they should take 3 credits from CAR M and 3 credits from specially-designed CAR A with English-Language and English Reading and Writing Requirements. They will also need to fulfil Chinese Reading and Writing requirements.

2.5.1.7 Healthy Lifestyle

Healthy lifestyle is the platform for all-round development. All students (except senior year students) are required to successfully complete a non-credit-bearing programme in healthy lifestyle. Students will be required to complete the following components:

(1) sports training/participation,
(2) e-learning modules, and
(3) lectures/talks.

The syllabus covers physical health, mental health, social health, spiritual health, values and priorities on healthy behaviour with reference to competing priorities in life, reflections on healthy living, and plans for self-improvement or maintaining healthy behaviour. Details of the programme can be found at: http://www.polyu.edu.hk/ogur/student/4yr/gur/hls/revised.

Students in Articulation Degree programmes and Senior Year intakes to the 4-year Ug degree programmes are not required to take the Healthy Lifestyle Programme. Advanced Standing students are required to take HLS (with the exception of those who are HD/AD holders following the Senior Year/Articulation Degree programme GUR curriculum).

2.5.1.8 Essential Components in General Education (for Senior year students only)

Senior year students need to complete a non-credit-bearing requirement in General Education via e-modules in their year 1 study. The Essential Components of General Education includes four modules namely Academic Integrity; AI and Data Analytics; Innovation and Entrepreneurship; and National Education.

2.6 Study Load and Subject Registration

2.6.1 Minimum and Maximum Credits

Students in the BSc (Hons) Scheme in Computing and AI are required to complete a ***minimum of 120 academic*** credits and ***4 WIE*** credits to graduate (*60 academic* credits and *4 WIE* credits for senior year students). They may complete a ***maximum of 150*** credits without having to pay extra tuition fees (*75* credits for senior year students).

Students (except senior year students) could take on an additional minor for 18 credits. If they wish, they are allowed to make use of the free subjects and appropriate CAR subjects within the 120 credits degree requirement to partially satisfy the minor requirement while staying within a total of 150 credits.

2.6.2 Study Load

For students following the progression pattern specified for their programme, they have to take the number of credits and subjects, as specified in the Programme Requirement Document, for each semester. Students cannot drop those subjects assigned by the department unless prior approval has been given by the department.

The normal study load is 15 credits in a semester for full-time study. The maximum study load to be taken by a student in a semester is 21 credits, unless exceptional approval is given by the Head of the programme offering Department. For such cases, students should be reminded that the study load approved should not be taken as grounds for academic appeal.

To help improve the academic performance of students on academic probation, these students will be required to take a reduced study load in the following semester (Summer Term excluded). The maximum number of credits to be taken by COMP students on academic probation is 12 (subject to review and change).

Students are not allowed to take zero subject in any semester, including the mandatory summer term as required by some programmes, unless they have obtained prior approval from the programme offering Department; otherwise they will be classified as having unofficially withdrawn from their programme. Students who have been approved for zero subject enrolment (i.e. taking zero subject in a semester) are allowed to retain their student status and continue using campus facilities and library facilities. Any semester in which the students are allowed to take zero subject will nevertheless be counted towards the total period of registration.

Students who have obtained approval to pace their studies and students on programmes without any specified progression pattern who wish to take more than the normal load of 15credits in a semester should seek advice from the Department concerned before the selection of subjects.

2.6.3 Subject Registration and Withdrawal

In addition to programme registration, students need to register for the subjects at specified periods prior to the commencement of the semester. An add/drop period will also be scheduled for each semester / term. Students may apply for withdrawal of their registration on a subject after the add / drop period, if they have a genuine need to do so. The application should be made to the relevant programme offering Department and will require the approval of both the subject teacher and the host Department Programme Leader concerned (or an alternate academic staff authorised by the programme offering Department). Applications submitted after the commencement of the examination period will not be considered. For approved applications of subject withdrawal, the tuition fee paid for the subject will be forfeited and the withdrawal status of the subject will be shown in the assessment result notification and transcript of studies, but will not be counted in the calculation of the GPA.

The pre-requisite requirements of a subject must have been fulfilled before a student registers for that subject. However, the subject offering Department has the discretion to waive the pre-requisite requirements of a subject, if deemed appropriate. If the pre-requisite subject concerned forms part of the requirements for award, the subject has to be passed in order to satisfy the graduation requirements for the programme concerned, despite the waiving of the pre-requisite.

Subject to the maximum study load of 21 credits per semester and the availability of study places, students are allowed to take additional subjects on top of the prescribed credit requirement for award before they become eligible for graduation. Students will be allowed to take additional subjects for broadening purpose, after they fulfil the graduation requirements and for the following semester. However, they will still be subject to the maximum study load of 21 credits per semester and the availability of places in the subjects concerned, and their enrolment will be arranged as subject-based students only and be subject to the rules on 'Admission of Subject-based Students', except that graduates from UGC-funded programmes will not be restricted to taking only subjects from a self-financed programme.

2.6.4 Credit Transfer

2.6.4.1 Students may be given credits for recognised previous studies including mandatory General University Requirements (GUR) subjects; and the credits will be counted towards meeting the requirements for award. Transferred credits may be counted towards more than one award. The granting of credit transfer is a matter of academic judgment. In assessing the transferability of subjects previously taken, the syllabus of that subject should be carefully scrutinized to ascertain that it is comparable to the PolyU's curriculum. Whether the previous studies are from institutions on credit-based or non-credit-based system should not be a matter of concern, and the subject size need not be a perfect match. To ascertain the academic standing of the institution offering the previous studies, the Department might need to request the institutions concerned to provide more relevant information.

2.6.4.2 Credit transfer may be done with or without the grade being carried over; the former should normally be used when the credits were gained from PolyU. Credit transfer with the grade being carried over may be granted for subjects taken from outside the University, if deemed appropriate, and with due consideration to the academic equivalence of the subjects concerned and the comparability of the grading systems adopted by the University and the other approved institutions. Subject credit transfer is normally decided by the subject offering Department. However, for applications which

are submitted by students who have completed an approved student exchange programme, the decision will be made by the programme offering Department in consultation with the subject offering Departments. As the application for credit transfer may involve subjects offered by more than one Department, the programme offering Department should coordinate and check whether the maximum limit for credit transfer for a student has been exceeded, and whether the student has fulfilled the residential requirement as stipulated in Section 2.5.1 above.

2.6.4.3 In case of disagreement between the programme offering Department and the subject offering Department, the two Faculty Deans/School Board Chairmen concerned will make a final decision jointly on the application. The validity period of credits previously earned, is 8 years after the year of attainment.

2.6.4.4 Normally, not more than 50% of the credit requirement for award may be transferable from approved institutions outside the University. For transfer of credits from programmes offered by PolyU, normally not more than 67% of the credit requirement for award can be transferred. In cases where both types of credits are being transferred (i.e. from programmes offered by PolyU and from approved institutions outside the University), not more than 50% of the credit requirement for award may be transferred. For students admitted to an Articulation Degree or Senior Year curriculum which is already a reduced curriculum, they will not be given credit transfer for any required GUR subjects, and they must complete at least 60 credits to be eligible for award. Students exceptionally admitted to an Articulation Degree or Senior Year curriculum before 2017/18 based on qualification more advanced than Associate Degree/Higher Diploma may be given credit transfer for the required GUR subjects if they had completed comparable components in their earlier studies. These students can take fewer than 60 credits for attaining the award. As from the 2017/18 intake cohort, all students admitted to an Articulation Degree or Senior Year curriculum, irrespective of the entry qualifications they held when applying for admission to the programmes, are required to complete at least 60 credits to be eligible for award.

If the credits to be transferred are part of a PolyU programme which is accredited by a professional body, the Department concerned should ensure that the transferred credits will also meet the requirement of the relevant professional body.

If a student is waived from a particular stage of study on the basis of advanced qualifications held at the time of admission, the student concerned will be required to complete fewer credits for award. For these students, the 'deducted' credits at admission stage will be counted towards the maximum limit for credit transfer when students apply for further credit transfer after their admission. This also applies to students admitted to an Articulation Degree or Senior Year curriculum when they claim further credit transfer after admission.

Notwithstanding the upper limits stipulated above, (and unless professional bodies stipulate otherwise) students may be given more credit transfer than these upper limits (e.g. upon completion of an exchange programme as mentioned below), subject to their satisfying the residential requirement.

Credit transfer can be applicable to credits earned by students through study at a non-local partner institution under an approved exchange programme. Students should, before they start the exchange programme, seek prior approval from the programme offering Department (who will consult the subject offering Departments as appropriate) on their study plan and credit transferability. As with all other credit transfer applications, the Departments concerned should scrutinise the syllabuses of the subjects which the

students are going to take at the partner institution, and determine their credit transferability based on academic equivalence with the corresponding subjects on offer at the PolyU, and the comparability of the grading systems adopted by PolyU and the partner institution. The transferability of credits, and the suitability for allowing grades to be carried over, must be determined and communicated to students before they start the exchange programme. In order to overcome the problems associated with subject-to-subject mappings, block credit transfer rather than subject-by-subject credit transfer can be given.

All credit transfers approved will take effect only in the semester for which they are approved. A student who applies for transfer of credits during the re-enrolment or the add/drop period of a particular semester will only be eligible for graduation at the end of that semester, even if the granting of credit transfer will immediately enable the student to satisfy the credit requirement for the award.

Regarding credit transfer for GUR subjects, the Programme Host Department is the approval authority at the time of admission to determine the number of GUR credits which an Advanced Standing student will be required to complete for the award concerned. Programme Host Departments should make reference to the mapping lists of GUR subjects, compiled by the Committee on General University Requirements (CoGUR), on the eligibility of the subjects which can qualify as GUR subjects. Applications for credit transfer of GUR subjects after admission will be considered, on a case-by-case basis, by the Subject Offering Department or Office of General University Requirements (OGUR)/Office of Service Learning (OSL), in consultation with the relevant Sub-committee(s) under CoGUR, as appropriate.

For credit transfer of retaken subjects, the grade attained in the last attempt should be taken in the case of credit transfer with grade being carried over. Students applying for credit transfer for a subject taken in other institutions are required to declare that the subject grade used for claiming credit transfer was attained in the last attempt of the subject in their previous studies. If a student fails in the last attempt of a retaken subject, no credit transfer should be granted, despite the fact that the student may have attained a pass grade for the subject in the earlier attempts.

Students should not be granted credit transfer for a subject which they have attempted and failed in their current study unless the subject was taken by the student as an exchange-out student in his/her current programme.

2.6.5 Subject Exemption

Students may be exempted from taking any specified subjects, including mandatory General University Requirements (GUR) subjects, if they have successfully completed similar subjects previously in another programme or have demonstrated the level of proficiency/ability to the satisfaction of the subject offering Department. Subject exemption is normally decided by the subject offering Department. However, for applications which are submitted by students who have completed an approved student exchange programme, the subject exemption is to be decided by the programme offering Department in consultation with the subject offering Departments. If students are exempted from taking a specified subject, the credits associated with the exempted subject will not be counted towards meeting the award requirements (except for exemptions granted at admission stage). It will therefore be necessary for the students to consult the programme offering Department and take another subject in order to satisfy the credit requirement for the award.

2.6.6 Award Requirement

The distribution and summary of *minimum* number of required credits for the individual programmes is shown below:

	Normal Intake	Senior Year Intake
University Requirements		
Language and Communication Requirements (English)	6 credits	- (only except those not meeting the equivalent standard of the Undergraduate Degree LCR)
Language and Communication Requirements (Chinese)	3 credits	
Leadership and Intra-Personal Development	3 credits	-
Service Learning	3 credits	3 credits
Freshmen Seminar	3 credits	-
Cluster Areas Requirement	12 credits	6 credits
China Studies Requirement	(3 of the 12 CAR credits)	(3 of the 6 CAR credits)
Programme Requirements		
Programme Compulsory	60 credits	30 credits
Programme Elective	18 credits	21 credits
Free Elective	12 credits	-
Total	**120 credits**	**60 credits**
Other Requirements		
Work Integrated Education	4 WIE credits	4 WIE credits
Healthy Lifestyle	0 credit	-

Remarks: For 61431-FFT Normal Intake students, the credit requirement for Cluster Areas Requirement and Free Elective would be 9 credits and 15 credits respectively.

2.6.7 Eligibility for Award

A student would be eligible for award if he satisfies all the conditions listed below:

(i) Accumulation of the requisite number of credits for the particular award, as defined in the Programme Requirement Document; and

(ii) Satisfying the residential requirement for at least 1/3 of the credits to be completed for the award he is currently enrolled, unless the professional bodies stipulate otherwise; and

(iii) Satisfying all the "compulsory" and "elective" requirements as defined in the Programme Requirement Document; and

(iv) Having a *Grade Point Average* (GPA) of 1.70 or above at the end of the programme; and

(v) Satisfying other requirements as stipulated in the Programme Requirement Document e.g. Work-integrated experience, and any other language requirements.

A student is required to graduate as soon as he satisfies all the above conditions for award. The student concerned is required to apply for graduation, in the semester in which he/she is able to fulfil all his/her graduation requirements, and after the add/drop period for that semester has ended.

For Students taking the Major/Minor option (not applicable for senior year intake)

Students taking the Major/Minor option will be considered for an award when they have satisfied the requirements for both the Major and Minor studies (i.e. having a GPA of 1.70 or above) and have also submitted an application for graduation. If the 18 credits taken for the Minor study can meet the requirements for a specific Minor, the Major students may apply to graduate with a specific Minor, in addition to their Major. Otherwise, students will graduate with a Major only.

Subject to approval by the Minor-offering Department, students may count up to 6 credits from their Major/GUR [including Language Communication Requirements (LCR) subjects at proficiency level] towards their chosen Minor. Nevertheless, students must take at least 6 credits from their chosen Minor programme in order to satisfy the residential requirement of their chosen Minor. In addition, to be eligible for the Major and Minor awards, the total number of credits taken by the students for their Major-Minor studies must not be lower than the credit requirement of the single discipline Major programme.

For Students taking the Double Majors option

Students are required to obtain an overall GPA of at least 1.70, in order to satisfy the requirement for graduation with Double Majors. They will not be allowed to graduate with one of the 2 Majors. The total credit requirement of Double Majors will depend on the degree of commonality between the two Majors. Apart from the 30 credits of GUR subjects, up to 1/3 of the DSR of the first Major which are common to the second Major can be double-counted towards the second Major.

2.6.8 Normal Duration for Completion of a Programme

Students should complete the programme within the normal duration of the programme as specified in the Programme Requirement Document. Those who exceed the normal duration of the programme will be de-registered from the programme unless prior approval has been obtained from relevant authorities. The study period of a student shall exclude deferment granted for justifiable reasons, and the semester(s) when the student has been approved to undertake internship. Any semester in which the students are allowed to take zero subject will be counted towards their total period of registration.

Students who have been registered for the normal duration of the programme may request extension of their studies for up to one year with the approval of the relevant Heads of Department. Applications for extension of study period beyond one year and up to two years will require the approval from Faculty Board Chairman.

Students who have exceeded the normal duration of the programme for more than two years and have been de-registered can submit an appeal to the Academic Appeals Committee to request further extension. If the appeal fails, the student shall be de-registered.

2.7 Admission to Minor and Double Majors; Declaration and Transfer (not applicable for senior year intake)

2.7.1 Programme Declaration and Transfer

The BSc (Hons) Scheme in Computing and AI is composed of these awards: BSc in Computing, BSc in Enterprise Information Systems, and BSc in Financial Technology and Artificial Intelligence.

Students admitted to this scheme programme should submit a form to the department to declare the target award within the scheme after semester 2 of their year 1 study. With approval from the department, students may transfer to another award within the scheme at a later stage of their study.

In addition, the 4-year undergraduate degree framework allows students to work for a single discipline Major, a Major plus a Minor (unless the Major is so designed as to preclude the possibility of a further Minor study), or a Double Majors.

2.7.2 Admission to Minor Programmes

Minor study will be a free choice by students and not mandatory. Normally, this option to study for a Minor will not be applicable to students who are admitted to the advanced stage of a programme, nor to students who are admitted to an articulation degree programme. Students who opt for Minor study will be subject to the following regulations.

(i) A Minor programme is a collection of subjects totalling 18 credits with at least 50% (9 credits) of the subjects at Level 3 or above. The subjects under a Minor should have a coherent theme introducing students to a focused area of study.

(ii) Students interested in a Minor must submit their applications to and obtain approval from the Minor-offering Department, at the start of second year of study. Students should submit their applications to their Major Department, which will indicate its support or otherwise, before the Minor-offering Department makes a final decision on the application;

(iii) Students are expected to complete their approved Minor as part of their graduation requirements. Students who wish to withdraw from a Minor need to apply for approval officially from the Minor offering department, before the end of the add/drop period of the last Semester of study;

(iv) Students with approved Minor will be given a higher priority in taking the Minor subjects over the students who take the subjects as free-electives; 'Free electives' refers to any subjects (including CAR subjects) offered by the University, unless otherwise specified;

(v) Subject to approval by the Minor-offering Department, students may count up to 6 credits from their Major/GUR [including Language Communication Requirement (LCR) subjects at proficiency level] towards their chosen Minor. Nevertheless, students must take at least 6 credits from their chosen Minor programme in order to satisfy the residential requirement of their chosen Minor. In addition, to be eligible for the Major and Minor awards, the total number of credits taken by the students for their Major-Minor studies must not be lower than the credit requirement of the single discipline Major programme;

(vi) Only students with a GPA of 2.5 or above can be considered for Minor study enrolment. The Minor-offering Department may set a quota and additional admission requirements for their Minor; and

(vii) Students are required to obtain a GPA of at least 1.70 in order to satisfy the requirement for graduation with a Major plus a Minor.

2.7.3 Admission to Double Majors

Double Majors will provide an opportunity for the more capable students, who are interested in expanding their study beyond a single degree, to take a Second Major study. Students who opt for a double Major study will be subject to the following regulations.

(i) Completion of Double Majors requires more than the normative study period of 4/5 years and extra credits on self-financed basis (i.e. higher tuition fee). The total credit requirements of a Double Major will depend on the degree of commonality between the 2 Majors. Apart from the 30 credits of GUR subjects, up to 1/3 of the Discipline-Specific Requirements (DSR) of the First Major which are common to the Second Major can be double-counted towards the Second Major.

(ii) Students who wish to take a Second Major must obtain approval from the host Department of the first Major. They can then submit their applications to the second Major-offering department starting from their second year of study. Only students with a GPA of 3.0 or above can be considered for admission to a Second Major, while Departments offering the Second Major can stipulate a higher GPA requirement if appropriate.

(iii) Students enrolled in a second Major will be given priority in taking second Major subjects over the students who take the subjects as free electives.

(iv) Students will be put on academic probation if they fail to obtain a GPA of 1.70 or above.

(v) Students are required to obtain an overall GPA of at least 1.70, in order to satisfy the requirement for graduation with Double Majors. They will not be allowed to graduate with one of the 2 Majors.

(vi) Students who wish to withdraw from a Second Major must obtain approval from the Department offering the Second Major, before the end of the add/drop period of the last Semester of study.

(vii) Students will not be allowed to withdraw from a First Major and continue with the Second Major only.

2.8 Teaching, Learning and Assessment

The specific learning outcomes expected to be achieved by a subject are spelt out explicitly in its syllabus. Students can know the aims and expectation of every subject before learning. They can also conduct a self-assessment to evaluate whether the specified specific learning outcomes of the subject have been achieved at the conclusion of the subject. Some of the specific learning outcomes as depicted in Section **1.6** are further elaborated into subject level learning outcomes according to the particular nature of a subject. The approaches used to achieve the specific learning outcomes of a subject include lectures, tutorials, seminars, laboratories, and case studies are described clearly in the syllabus of the subject. In line with the PolyU Strategic Plan (2019/20 - 2024/25), innovative and effective educational technologies and methodologies will also be used to enhance student learning, such as strengthening student ability to "learn to learn", using public domain learning materials to learn from the best, setting up peer learning groups, making more extensive use of interactive pedagogies and employing learning analytics.

The prime purpose of assessment is to enable students to demonstrate that they have met the aims and objectives of the academic programme in particular that they have fulfilled the requirement of each subject and have, at the end of their study achieved the standard appropriate to the award. In the development of the curriculum, we have ensured that every teaching and learning approach should be assessed with the most appropriate assessment method, inclusive of examination, test, quiz, project and case-study report and presentation, assignment, laboratory work and programming exercise and so on. Since it is important to assess student performance in a holistic manner, both continuous assessment component and examination component are adopted.

Assessment should fulfil two major functions. It is used to evaluate whether the specific learning outcomes of a subject have been achieved by the students, and to distinguish their performance in achieving them. The criteria-referenced assessment approach is applied and assessment should not be made on a relative basis with reference to a norm. Students' performance in a subject will be assessed by "how much" and "how good" that the specific criteria as specified in its syllabus can be achieved. In case of group activity, such as the capstone project, both the overall performance of the group as well as individual effort/contribution of each team member should be clearly assessed. Assessment also serves as a prompt and useful feedback to students. Students will be informed of their performance in the assessment so that they are aware of their progress and attainment on a continuous basis to facilitate teaching and learning.

Students' performance in a subject can be assessed by continuous assessment and/or examination as deemed appropriate. Where both continuous assessment and examinations are used, the weighting of each in the overall subject grade will be clearly stated in the subject syllabus. Continuous assessment may include tests, assignments, projects, laboratory work, field exercises, presentations and other forms of classroom participation. At the beginning of each semester, the subject teacher would inform students of the details of the methods of assessments to be used. Normally, the weighting of the examination is below 50% for most of the subjects offered by the Department. A few subjects adopt a 100% continuous assessment arrangement.

2.9 General Assessment Regulations (GAR)

Academic regulations for assessment, progression, de-registration, re-assessment, subject retake, subject registration, study deferment, subject exemption, and credit transfer, etc. are published in the ***Student Handbook***.

2.9.1 Assessment Methods

Students' performance in a subject shall be assessed by coursework and/or examinations and/or continuous assessment as depicted in individual subject syllabus. Coursework may include tests, assignments, projects, laboratory work, field exercises, presentations and other forms of classroom participation. The contribution made by each student in coursework involving a group effort shall be determined and assessed separately.

Assessment grades shall be awarded on a *criterion-referenced basis*. At the beginning of each semester, the subject teacher will inform students of the details of the methods of assessments to be used within the assessment framework. Criteria would be given for each subject with respect to the expectation. Performance in the Coursework and/or Final Examination components will be assessed, according to the weighting factor of both components in a certain subject.

The grade conversion of assessment result is as specified in the General Assessment Regulations, according to the following table:

Subject grade	***Short description***	***Elaboration on subject grading description***
A+ A A-	Excellent	Demonstrates excellent achievement of intended subject learning outcomes by being able to skillfully use concepts and solve complex problems. Shows evidence of innovative and critical thinking in unfamiliar situations, and is able to express the synthesis or application of ideas in a logical and comprehensive manner.
B+ B B-	Good	Demonstrates good achievement of intended subject learning outcomes by being able to use appropriate concepts and solve problems. Shows the ability to analyse issues critically and make well-grounded judgements in familiar or standard situations, and is able to express the synthesis or application of ideas in a logical and comprehensive manner.
C+ C C-	Satisfactory	Demonstrates satisfactory achievement of intended subject learning outcomes by being able to solve relatively simple problems. Shows some capacity for analysis and making judgements in a variety of familiar and standard situations, and is able to express the synthesis or application of ideas in a manner that is generally logical but fragmented.
D+ D	Pass	Demonstrates marginal achievement of intended subject learning outcomes by being able to solve relatively simple problems. Can make basic comparisons, connections and judgments and express the ideas learnt in the subject, though there are frequent breakdowns in logic and clarity.

Subject grade	*Short description*	*Elaboration on subject grading description*
F	Fail	Demonstrates inadequate achievement of intended subject learning outcomes through a lack of knowledge and/or understanding of the subject matter. Evidence of analysis is often irrelevant or incomplete.

'F' is a subject failure grade, whilst all others ('D' to 'A+') are subject passing grades. No credit will be earned if a subject is failed.

Notes:
- Marking rubrics aligned with these Grade Descriptors need not include all aspects of the grade descriptor.
- Marking rubrics aligned with these Grade Descriptors may include other aspects aligned with particular subject matter or field of study requirements but are not included in the grade descriptor.

Indicative descriptors for modifier grades

Main Grade (solid)	The student generally performed at this level, indicating mastery of the subject intended learning outcomes at this level.
+ (exemplary)	The student consistently performed at this level and exceeded the expectations of this level in some regards, but not enough to claim mastery at the next level.
- (marginal)	The student basically performed at this level, but the performance was inconsistent or fell slightly short in some regards.

Note: The above indicative descriptors for modifier grades are not applicable to the pass grades D and D+.

The grade points assigned to subject grades attained by students from 2020/21 are as follows:

Grade	*Grade Point for grades attained from 2020/21*
A+	4.3
A	4.0
A-	3.7
B+	3.3
B	3.0
B-	2.7
C+	2.3
C	2.0
C-	1.7
D+	1.3
D	1.0
F	0.0

A numeral grade point is assigned to each subject grade for the purpose of computation of the GPA. In the event that grade is awarded to subject components, a grade point with the decimal value may be generated for the overall result of the subject. This decimal-valued grade point will be converted to a letter grade. The corresponding grade point for the subject overall grade, instead of the actual grade points obtained by students, will be used for GPA calculation. This methodology for deriving subject overall grades only serves as an aid to subject assessors. As assessment should be a matter of judgement, not merely a result of computation, the subject lecturer will have the *discretion* to assign a grade which is considered to reflect more appropriately the overall performance of the student in a subject to override the grade derived by the computer. As students are to be

evaluated in a *holistic* manner, the Department does not require students to pass ***both*** continuous assessment and examination components in order to pass a subject.

All assessment grades proposed by subject lecturers must be endorsed by the respective Subject Assessment Review Panel for subjects offered by individual departments.

2.9.2 Retaking of Subjects

Students may only retake a subject which they have failed (i.e. Grade F or S or U). Retaking of subjects is with the condition that the maximum study load of 21 credits per semester is not exceeded.

The number of retakes of a subject should be restricted to two, i.e. a maximum of three attempts for each subject is allowed.

In cases where a student takes another subject to replace a failed elective subject, the fail grade will be taken into account in the calculation of the GPA, despite the passing of the replacement subject. Likewise, undergraduate or sub-degree students who fail a Cluster Area Requirement (CAR) subject may need to take another subject from the same Cluster Area in order to fulfil this part of the GUR, since the original CAR subject may not be offered; in such cases, the fail grade for the first CAR subject will be taken into account in the calculation of the GPA, despite the passing of the second CAR subject*.

Students need to submit a request to the Faculty/School Board for the second retake of a failed subject.

Students who have failed a compulsory subject after two retakes and have been de-registered can submit an appeal to the Academic Appeals Committee (AAC) for a third chance of retaking the subject.

In relation to above, in case AAC does not approve further retakes of a failed compulsory subject or the taking of an equivalent subject with special approval from the Faculty, the student concerned would be de-registered and the decision of the AAC shall be final within the University.

* In these circumstances when students do not have a choice to retake a failed subject, such as when the failed subject has been phased out, a 'tie-subject' arrangement can be made with the approval of the Faculty/School Board. Under the arrangement, another appropriate subject can be taken as equivalent to the subject which is not offered. Upon passing the equivalent subject, the fail grade of the original subject will be replaced by the latest grade of the retake subject and the failure grade of the original subject will not be taken into account in the calculation of the GPA.

2.9.3 GPA Calculations

At the end of ***each semester/term***, a Grade Point Average (GPA) will be computed as follows, and based on the grade point of all the subjects:

$$\text{GPA} = \frac{\sum_{n=1}^{N} \text{Subject Grade Point}_n \times \text{Subject Credit Value}_n}{\sum_{n=1}^{N} \text{Subject Credit Value}_n}$$

where N = number of all subjects (inclusive of failed subjects) taken by the student up to and including the latest semester/term. For subjects which have been retaken, only the grade point obtained in the final attempt will be included in the GPA calculation

The following subjects will be excluded from the GPA calculation:

(i) Exempted subjects
(ii) Ungraded subjects
(iii) Incomplete subjects
(iv) Subjects for which credit transfer has been approved, but without any grade assigned
(v) Subjects from which a student has been allowed to withdraw (i.e. those with the code 'W')

Subject which has been given an "S" code (i.e. absent from all assessment components) will be included in the GPA calculation and will be counted as "zero" grade point. GPA is thus the **unweighted** cumulative average calculated for a student, for all relevant subjects taken from the start of the programme to a particular point of time. GPA is an indicator of overall performance, and ranges from 0.00 to 4.30 from 2020/21.

The Weighted GPA will be used as a guide to help determine award classifications. For calculating the weighted GPA (and award GPA) to determine the Honours classification of students who satisfy the graduation requirements of Bachelor's degree award, a University-wide standard weighting will be applied to all subjects of the same level, with a weighting of 2 for Level 1 and 2 subjects, a weighting of 3 for Level 3 and 4 subjects. Same as for GPA, Weighted GPA ranges from 0.00 to 4.30 from 2020/21.

Weighted GPA will be computed as follows:

$$\text{Weighted GPA} = \frac{\sum_{n=1}^{N} \text{Subject Grade Point}_n \times \text{Subject Credit Value}_n \times W_n}{\sum_{n=1}^{N} \text{Subject Credit Value}_n \times W_n}$$

where W_n = weighting to be assigned according to the level of the subject

N = number of all subjects counted in GPA calculation as set out above.

Any subjects passed after the graduation requirement has been met or subjects taken on top of the prescribed credit requirements for award will not be taken into account of in the grade point calculation for award classification. However, if a student attempts more elective subjects (or optional subjects) than those required for graduation in or before the semester in which he/she becomes eligible for award, the elective subjects (or optional

subjects), except for subjects which are selected by students to fulfil the free electives requirement for graduation, with a higher grade/contribution shall be included in the grade point calculation (i.e. the excessive subjects attempted with a lower grade/contribution, including failed subjects, will be excluded).

2.9.4 Progression/Academic Probation/Deregistration

The Board of Examiners (BoE) shall, at the end of each semester (except for Summer Term unless there are students who are eligible to graduate after completion of Summer Term subjects), determine whether each student is:

(a) eligible for progression towards an award; or
(b) eligible for an award; or
(c) required to be deregistered from the programme.

When a student has a Grade Point Average (GPA) lower than 1.70, he/she will be put on academic probation in the following semester. If a student is able to pull his/her GPA up to 1.70 or above in the following semester, the status of "academic probation" will be lifted. The status of "academic probation" will be reflected in the examination result notification but not in the transcript of studies.

A student will have 'progressing' status unless he/she falls within the following categories, either of which may be regarded as grounds for deregistration from the programme:

(1) the student has reached the final year of the normal period of registration for that programme, as specified in the Programme Requirement Document, unless approval has been given for extension; or
(2) the student has reached the maximum number of retakes allowed for a failed compulsory subject; or
(3) the student's GPA is lower than 1.70for two consecutive semesters and his/her Semester GPA in the second semester is also lower than 1.70; or
(4) the student's GPA is lower than 1.70 for three consecutive semesters.

When a student falls within the categories as stipulated above, except for category (1) with approval for extension, the Board of Examiners shall de-register the student from the programme without exception.

A student may be deregistered from the programme enrolled before the time frame specified in the above conditions (3) or (4) if his/her academic performance is poor to the extent that the Board of Examiners (BoE) deems that his/her chance of attaining a GPA of 1.70 at the end of the programme is slim or impossible.

The progression of students to the following academic year will not be affected by the GPA obtained in the Summer Term, as Summer Term study is not mandatory for all students of the programme.

If the student is not satisfied with the de-registration decision of the Board of Examiners, he/she can lodge an appeal. All such appeal cases will be referred directly to Academic Appeals Committee (AAC) for final decision. Views of Faculties/Schools/Departments will be sought and made available to AAC for reference.

2.9.5 Classification of Awards

The following are guidelines for Boards of Examiners' reference in determining award classifications:

Honours degrees	*Guidelines*
1st	The student's performance/attainment is outstanding, and identifies him/her as exceptionally able in the field covered by the programme in question.
2:i	The student has reached a standard of performance/ attainment which is more than satisfactory but less than outstanding.
2:ii	The student has reached a standard of performance/ attainment judged to be satisfactory, and clearly higher than the "essential minimum" required for graduation.
3rd	The student has attained the "essential minimum" required for graduation at a standard ranging from just adequate to just satisfactory.

Under exceptional circumstances, a student who has completed an Honours degree programme, but has not attained Honours standard, may be awarded a Pass-without-Honours degree. A Pass-without-Honours degree award will be recommended, when the student has demonstrated a level of final attainment which is below the 'essential minimum' required for graduation with Honours from the programme in question, but when he/she has nonetheless covered the prescribed work of the programme in an adequate fashion, while failing to show sufficient evidence of the intellectual calibre expected of Honours degree graduates. For example, if a student in an Honours degree programme has a Grade Point Average (GPA) of 1.70 or more, but his/her Weighted GPA is less than 1.70, he/she may be considered for a Pass-without-Honours classification. A Pass-without-Honours is an unclassified award, but the award parchment will not include this specification.

Students who have committed academic dishonesty or non-compliance with examination regulations will be subject to the penalty of the lowering of award classification by one level. For undergraduate students who should be award a Third class Honours degree, they will be downgraded to a Pass-without-Honours. The minimum of downgraded overall result will be kept at a pass. In rare circumstances where both the Student Discipline Committee and Board of Examiners of a Department consider that there are strong justifications showing the offence be less serious, the requirement for lowering the award classification can be waived.

The following are the award GPA ranges for determining award classifications:

Honours classification	*Award GPA*
1st	3.60 – 4.30
2:i	3.00 – 3.59
2:ii	2.40 – 2.99
3rd	1.70 – 2.39

Major/Minor Study

For students who have completed a Major/Minor programme, a single classification will be awarded and their award classification will mainly be based on the "Major GPA", but it can be moderated by the Board of Examiners with reference to the "Minor GPA". For students who have completed a Major programme combined with free electives, their award classification will be determined by their "Major GPA" which includes grades obtained for the free electives, if appropriate.
"Major GPA" is derived based on all subjects of the Major programme, including those meeting the University's mandatory general education requirement and programme-specific language requirement.

"Minor GPA" is derived based on the 18 credits of Minor study (either a specific Minor or free combination of electives). Level weighting will not be included in the calculation of Minor GPA.

Double Majors

The derivation of GPA for award classification for the First Major and Second Major will be decided by the Department offering the Major programme. Students will be given two award parchments, one for each Major programme, which will be issued upon completion of both Majors. The honours classification of the two Major awards need not be identical.

2.9.6 Exceptional Circumstances

Absence from an assessment component

If a student is unable to complete all the assessment components of a subject due to illness or other circumstances beyond his/her control, and considered by the Subject Assessment Review Panel (SARP) as legitimate, the Panel will determine whether the student will have to complete the assessment and, if so, by what means. This assessment shall take place at the earliest opportunity, and before the commencement of the following academic year.

The student concerned is required to submit his/her application for late assessment in writing to the Head of Department offering the subject, within five working days from the date of the examination, together with any supporting documents. Approval of applications for late assessment and the means for such late assessments shall be given by the SARP.

Assessment to be completed

For cases where student fail marginally in one of the components within a subject, the BoE can defer making a final decision until the students concerned have completed the necessary remedial work to the satisfaction of the subject examiner(s). The remedial work must not take the form of re-examination.

Aegrotat award

If a student is unable to complete the requirements of the programme for the award due to very serious illness, or other very special circumstances which are beyond his/her control, and considered by the Board of Examiners (BoE) as legitimate, the Faculty Board will determine whether the student will be granted an aegrotat award. An aegrotat award is normally granted only under very exceptional circumstances.

A student who has been offered an aegrotat award shall have the right to opt either to accept such an award, or request to be assessed on another occasion to be stipulated by the BoE. The student's exercise of this option shall be irrevocable. The acceptance of an aegrotat award by a student shall disqualify him from any subsequent assessment for the same award.

An aegrotat award shall normally not be classified, and the award parchment shall not state that it is an aegrotat award. However, the BoE may determine whether the award should be classified provided that they have adequate information on the students' academic performance.

Other particular circumstances

A student's particular circumstances may influence the procedures for assessment but not the standard of performance expected in assessment.

2.9.7 Plagiarism

Plagiarism is a very serious matter. All students should read the session on "*Plagiarism*" which is provided in the ***Student Handbook***. The use of the ***turnitin*** plagiarism detection system to deter plagiarism is strongly encouraged by the University and is adopted in a number of subjects.

Students should also refer to the following *departmental policy* on plagiarism:

First offence: zero marks on the plagiarised assignment/ continuous assessment for both the copycat and the one who offers to be copied **and** deduct additional marks that carries the same weight as the assessed component. For example, if a student is caught plagiarising on a written assignment which contribute to 20% of the total grade, both the copycat and the one who offer his/her assignment to be copied will receive zero marks for 40% (i.e. 20% X 2) of the grade. For serious first offences, teaching staff have the discretion to impose the penalty for serious or repeated offences.

For serious or repeated offences: the case will be submitted to the department's Student Discipline Task Group or the University's Student Discipline Committee for consideration and decision.

2.9.8 Recording of Disciplinary Actions in Students' Records

With effect from Semester One of 2015/16, disciplinary actions against students' misconducts will be recorded in students' records.

Students who are found guilty of academic dishonesty or non-compliance with examination regulations will be subject to the penalty of having the subject result concerned disqualified and be given a failure grade with a remark denoting 'Disqualification of result due to academic dishonesty/non-compliance with examination regulations'. The remark will be shown in the students' record as well as the assessment result notification and transcript of studies, until their leaving the University.

Students who have committed disciplinary offences (covering both academic and non-academic related matters) will be put on 'disciplinary probation'. The status of 'disciplinary probation' will be shown in the students' record as well as the assessment result notification, transcript of studies and testimonial during the probation period, until

their leaving the University. The disciplinary probation which is normally one year unless otherwise decided by the Student Discipline Committee.

The University reserves the right to withhold the issuance of any certification of study to a student/graduand who has unsettled matters with the University, or is subject to disciplinary action.

Part 3: Programme Management, Resource and Support

3.1 Programme Operation and Management

The BSc (Hons) Scheme in Computing and AI is managed by the *Departmental Undergraduate Programme Committee* stipulated by guidelines set by the University. This committee will be responsible for the academic standards, content, delivery and assessment of all undergraduate programmes within its purview. Specifically, the Departmental Undergraduate Programme Committee will comprise:

- Head of Department;
- Departmental Learning and Teaching Committee Chair;
- Scheme Leader and Programme leader of each streams;
- Representatives of subject lecturers from each programme;
- Representatives from co-hosting department for each double-degree programme;
- Student representatives from each programme.

A small Programme Execution Group will manage the day-to-day operation of each constituent programme under the BSc (Hons) Scheme in Computing and AI. The group will operate in an informal manner, being organized by the Award Coordinator/Programme Leader with approval from Head of Department. The Programme Execution Groups for the programmes report to the Departmental Undergraduate Programme Committee.

The programme gets feedback from students through the Student/Staff Consultative Group, which is a window for students to contribute to better the programme. Two student representatives from each year are included from each award to participate in the student/staff consultative group. The group consists of the Award Coordinator/Programme Leader, subject lecturers and student representatives. It will meet approximately twice a year for constructive discussion of the award and the programme and of possible improvements, in a less formal context than the Programme Committee. This group is comprised of:

- Programme Leader
- Subject Leaders
- Student representatives (2 to 3 from each year).

Other committees relevant to the management of the programme include:

3.1.1 Departmental Learning and Teaching Committee

The Departmental Learning and Teaching Committee (DLTC) will be overseeing the operation and administration of all the awards hosted by the Department of Computing. It is formed under the regulations of the department and the university. Its main duties are:

- To promote quality learning and teaching in the department, particularly at the classroom level, and at the teacher-student interface.
- To keep under review the quality of learning and teaching in the department.
- To develop strategies and guidelines relating to the enhancement of learning and teaching quality in the department.
- To evaluate and prioritize proposals from departmental staff on learning and teaching development projects - for funding by the DLTC and other relevant agencies.
- To monitor progress of learning and teaching development projects carried out in the department.

3.1.2 Subject Assessment Review Panel (SARP)

The Department's *Subject Assessment Review Panel* (SARP) will review the distribution of grades within a subject and finalize the grades at the end of each semester before submission to BoE. The SARP shall include the Head of the Department offering the subjects and acts as the Chairman and the relevant subject examiners. It is also responsible for deciding upon the granting of re-assessment to students and the form of such re-assessment.

3.1.3 Board of Examiners (BoE)

A *Board of Examiners* (BoE) of the Programme will meet at the end of each semester after the grades are endorsed by the SARP to consider students' classifications of award and to deal with problematic cases. The BoE will consider and make decisions concerning the progression, academic probation, deregistration cases of students. In addition, it will decide on the eligibility for awards and award classification for graduating students.

3.2 Facilities and Support

The Department attaches importance to the practical work of students. Academic programmes and research activities are well supported with a wide range of computing facilities available in the departmental Computing Laboratories (located at 5/F and 6/F of PQ Wing, and 5/F of P Wing) and the University's Information Technology Services Office (ITS) via the departmental Local Area Network (LAN). The departmental LAN consists mainly of 10Gbps Ethernet segments interconnected together using the state-of-art network switches, which are provided and maintained by the University.

The departmental LAN is also connected to The Hong Kong Polytechnic University campus Gigabit Network and then to the Internet. The Internet connection is used for web access, electronic mail, internet news, remote logins, file transfers, and other forms of interaction with the world-wide computing community.

A wide variety of computing hardware and software is available to support both the undergraduate and postgraduate teaching. They are listed in Section **3.2.1** and **3.2.2**. While some of the computing facilities mentioned in the list are located in the offices of academic staff and research staff, most of them are located in the departmental Computing Laboratories as depicted in Section **3.2.3**.

3.2.1 Computing Hardware

- 9 sets of Dell PowerEdge R620 Server
- 23 sets of IBM X3550 Servers
- 400 sets of Intel i7 PCs with 16GB memory
- 115 sets of Intel i7 PCs with 32GB memory and Nvidia GeForce 1080 GPU card
- 25 sets of Intel i9 PCs with 64GB memory and Nvidia GeForce 1080Ti GPU card
- 5 sets of Intel i7 PCs with 32GB memory and Nvidia GeForce 3070 GPU card
- 7 sets of Xerox Multifunction Machines (Print/Copy/Scan)

- 3 sets of Xerox Colour Multifunction Machines (Print/Copy/Scan)

3.2.2 Computing Software

- Data mining and warehousing tool
  - SAS Enterprise Data Miner and Text Miner
  - SPSS Data Mining (PASW)
- Database management system
  - Oracle
  - Microsoft SQL Server
  - MySQL Server
- Distributed computing software
  - MPI
  - Hadoop
- Graphics and Game Development tools
  - Unity
  - 3D Studio Max
  - Microsoft Visio
- Office software
  - Microsoft Office
- Operation system software
  - Microsoft Window 10
  - Microsoft Window server
  - Ubuntu Linux
  - CentOS Linux
- Programming language
  - Java
  - Microsoft Visual Studio.Net
  - Anaconda suite with Python / Tensorflow / PyTorch
- Simulation packages
  - CSIM
  - FlexSim
- Software Engineering / Project Management packages
  - Microsoft Project
- Statistical and Mathematical analysis tools
  - Matlab
  - SAS
  - SPSS
- Web Development
  - Adobe Creative Suite
  - Microsoft Expression
  - Apache/Tomcat

3.2.3 Computing Laboratories

PC Laboratories (4 rooms)

- are located on 6/F of PQ Wing
- provide the facilities for basic and advanced programming on the Window environment for all students in the department;
- serve as an instruction laboratory with overhead LCD projectors during some class hours;
- are supported by a total of 105 sets of Intel i7 PCs with 32GB memory and Nvidia GeForce 1080 GPU, and 25 sets of Intel i9 PCs with 64GB Memory and Nvidia GeForce 1080Ti GPU card.

Project Laboratory

- is located on 5/F of P Wing
- is dedicated to the final year Undergraduate and Postgraduate students, who carry out their final year project implementations;
- has Wireless Access points for notebook PC connections to the campus network;
- is supported by 5 sets of Intel i7 PCs with 32GB memory and Nvidia GeForce 1080 GPU.

Student Laboratory (2 rooms)

- are located on both 6/F of PQ Wing and 5/F of P Wing
- provides a general computing environment for student's work;
- has Wireless Access points for user's notebook PC connection to campus network;
- is supported by a total of 5 sets of Intel i7 PCs with 32GB memory and Nvidia GeForce 1080 GPU, plus 4 sets of high speed laser printers.

Research / Teaching laboratories

A number of special laboratories including:

- Biometrics Research and Innovation Centre
- Data Science and AI Laboratory
- FinTech and Cyber Security Laboratory
- Internet and Mobile Computing Laboratory
- PolyU-Microsoft Smart Computing Laboratory
- PolyU-Yonyou Joint Laboratory on Smart Cloud Computing

have been created for research development and teaching support.

3.2.4 Teaching and Learning Support through a Learning Management System

To enhance interactive learning and facilitate communications, the proposed programme will be supported by a learning management system (LMS). The University is using a BlackBoard-based LMS called LEARN@PolyU. The LMS is a web-based teaching and online content management system. All the essential functions for interactive teaching/learning through Internet are built-in as standard features of the LMS. These include:

1. Online Bulletin Board (newsgroup)
2. Online Chat Room (similar to ICQ)
3. Online Whiteboard (for drawing pictures jointly through Internet)
4. Web Page Creation (with What You See Is What You Get tools)
5. Interactive Quiz (with What You See Is What You Get tools)
6. Internal Mail
7. Class Calendar
8. Internal Content Search
9. Password Authentication
10. Student database for class management and performance tracking

Furthermore, multimedia components such as digital video, digital sound, computer graphics and animations can be imported into LMS for creating web-based lessons.

Part 4: Subject Syllabi

The department reserves the rights to revise and update the syllabi whenever appropriate and deems necessary.

Individual subject syllabus could be view at https://www38.polyu.edu.hk/ePublic/subject-search.jsf.